

Received 1 February 2025, accepted 22 April 2025, date of publication 30 April 2025, date of current version 7 May 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3565812

RESEARCH ARTICLE

Anomaly Detection in Industrial Machine Sounds Using High-Frequency Features and Gate Recurrent Unit Networks

THI-THU-HUONG LE¹, ANDRO APRILA ADIPUTRA², JIWON YUN²,
AND HOWON KIM², (Member, IEEE)

¹Blockchain Platform Research Center, Pusan National University, Busan 609735, South Korea

²School of Computer Science and Engineering, Pusan National University, Busan 609735, South Korea

Corresponding authors: Thi-Thu-Huong Le (huongle@pusan.ac.kr) and Howon Kim (howonkim@pusan.ac.kr)

This work was supported in part by the Ministry of Science and ICT (MSIT), South Korea, through the Convergence Security Core Talent Training Business (Pusan National University) Support Program, supervised by the Institute for Information and Communications Technology Planning and Evaluation (IITP), under Grant RS-2022-II221201; and in part by MSIT through the Information Technology Research Center (ITRC) Support Program, supervised by IITP, under Grant RS-2020-II201797.

ABSTRACT Detecting anomalies in industrial sound is critical for maintaining operational efficiency, preventing costly equipment failures, and ensuring workplace safety. However, it presents significant challenges due to the complexity and variability of industrial environments, including background noise and fluctuating operating conditions. This paper proposes a comprehensive approach that leverages machine learning (ML) and deep learning (DL) techniques to address these challenges. Using three datasets, the Malfunctioning Industrial Machine Investigation and Inspection for Domain Generalization 2022 (MIMII DG 2022), the Detection and Classification of Acoustic Scenes and Events (DCASE) 2022, and DCASE 2024, we evaluate the performance of various ML and DL models under different experimental conditions. Our study focuses on feature extraction methods such as Mel-spectrograms and Mel-frequency Cepstral Coefficients (MFCCs) to capture critical acoustic characteristics of industrial machinery. Both supervised ML and DL techniques are employed to explore effective anomaly detection strategies. Extensive experimentation and evaluation using metrics such as confusion matrix, accuracy, precision, recall, and F1 score highlight the effectiveness of our approach in real-world industrial scenarios. Experimental results demonstrate that eXtreme Gradient Boosting (XGBoost) outperforms Support Vector Machine (SVM) and Decision Tree (DT) models in the ML approach across both feature sets. In the DL approach, Gated Recurrent Units (GRU) perform better on MFCC features than Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks. GRU emerges as the best-performing model on testing datasets, achieving a precision of 99.56% and an F1 score of 99.55% on the DCASE 2024 dataset, a precision of 94% and an F1 score of 93.95% on the DCASE 2022 dataset, and a precision of 92.2% and an F1 score of 92.06% on the MIMII DG 2022 dataset. These results underscore the potential of DL for real-time industrial sound analysis in predictive maintenance, offering significant improvements over traditional methods.

INDEX TERMS Anomalous sound detection, supervised learning, high-frequency feature, Mel-Spectrogram, Mel-frequency cepstral coefficients, gated recurrent units.

The associate editor coordinating the review of this manuscript and approving it for publication was Mohammad Zia Ur Rahman^{1b}.

I. INTRODUCTION

In contemporary industrial settings, efficient machinery operation is essential for maintaining productivity, ensuring workplace safety, and preventing costly equipment failures [1], [2], [3]. Anomalous sounds emitted by industrial equipment

often indicate potential malfunctions or mechanical failures, making industrial sound analysis a crucial component of proactive maintenance and fault diagnosis strategies [4], [5]. Recent advancements in signal processing and machine learning (ML) have significantly enhanced the analysis of physiological signals, with numerous applications in healthcare. For instance, Pachori et al. explored the detection of atrial fibrillation from PPG sensor data using Variational Mode Decomposition, improving diagnostic accuracy in cardiovascular health [6]. In another study, Pachori and Gandhi employed an FBSE-based approach to effectively distinguish seizure and normal EEG signals, demonstrating the potential of advanced feature extraction methods in neurological monitoring [7]. Furthermore, their work on automated human emotion recognition through EEG sub-bands, using the TQWT technique, highlights the growing intersection of signal processing and affective computing [8]. These studies reflect the evolving landscape of automated systems for analyzing biomedical signals, an area further explored in the introduction to cardiovascular signals by Pachori et al. (2024), where they discuss the integration of signal processing techniques with ML for enhanced cardiovascular data analysis [9]. This growing body of work underscores the critical role of innovative signal processing methodologies in improving diagnostic and monitoring systems across various domains.

However, anomaly detection in industrial sound poses several challenges. Industrial environments are characterized by their complexity and variability, factors such as background noise, varying operating conditions, and the presence of multiple types of machinery that contribute to the difficulty of identifying abnormal sound patterns [10], [11]. Furthermore, the high-dimensional and sequential nature of sound data further complicates the task, as traditional anomaly detection techniques may struggle to capture relevant features and temporal dependencies effectively [12], [13].

Motivated by these challenges and opportunities, this paper presents a novel approach to anomaly detection in industrial sound analysis. Our goal is to develop a comprehensive framework capable of accurately identifying anomalies in sound signals, thereby facilitating proactive maintenance and fault diagnosis in industrial settings. Using state-of-the-art ML and deep learning (DL) techniques, we aim to overcome the limitations of existing methods and provide insights into effective strategies for anomaly detection.

Our approach uses supervised ML and DL techniques, allowing us to exploit the strengths of both paradigms. Using three diverse datasets, the MIMII DG, the DCASE 2022, and the DCASE 2024, we evaluate the performance of various ML and DL models in different industrial contexts. Our study focuses on feature extraction methods such as Mel-Spectrogram and Mel-frequency cepstral coefficients (MFCCs), which are tailored to capture the relevant acoustic characteristics of industrial machinery.

The primary contribution of this work lies in the development and evaluation of a comprehensive anomaly detection

framework tailored specifically to industrial sound analysis. We propose a hybrid approach that approaches representative ML algorithms and DL models, allowing us to leverage the strengths of both paradigms. Furthermore, we investigate the effectiveness of various feature extraction methods, including Mel-Spectrogram and Mel-frequency cepstral coefficients (MFCCs), for capturing relevant acoustic characteristics in industrial sound data. Through extensive experimentation with real-world datasets, we demonstrate the effectiveness and scalability of our approach across different industrial scenarios.

The contributions of this paper include the following.

- A comprehensive evaluation of supervised ML and DL techniques for detecting anomalous sounds in industrial environments. The effectiveness of these models was assessed using three distinct industrial sound datasets, with evaluation metrics such as confusion matrix, accuracy, precision, recall, and F1 score.
- An investigation into feature extraction methods, focusing on a hybrid approach that combines high-frequency features, particularly Mel-Spectrogram and MFCCs, to improve anomaly detection accuracy.
- Experimental validation and performance analysis, demonstrating the effectiveness of the proposed approach for industrial anomaly detection applications.

This paper addresses the pressing need for robust anomaly detection techniques in industrial sound analysis. By combining advanced ML and signal processing techniques, we aim to provide practitioners with powerful tools for proactive maintenance and fault diagnosis in industrial settings. The remainder of this paper is organized as follows. Section II provides an overview of related work in anomaly detection and industrial sound analysis. Section III presents the methodology adopted in our study, including dataset descriptions, feature extraction techniques, and model architectures. Section V presents the experimental results and performance evaluation of the ML and DL models in the MIMII DG, DCASE 2022, and DCASE 2024 datasets. Furthermore, this section discusses the findings and implications of our study, highlighting the strengths and limitations of the proposed approach. Finally, Section VII concludes the paper with a summary of key findings and suggestions for future research directions.

II. LITERATURE REVIEW

The related work provides a comprehensive review of previous research efforts in anomaly detection and industrial sound analysis, shedding light on methodologies, techniques and findings relevant to the current study. Traditional ML (ML) algorithms, such as Support Vector Machines (SVM), Decision Trees (DT), and extreme gradient boosting (XGBoost), have been widely employed for anomaly detection tasks in industrial settings due to their ability to classify data based on learned patterns. Notably, studies like [14], [15], [16], [17], [18], and [19] successfully utilized SVM models to detect anomalies in vibration signals from rotating

TABLE 1. Comparative analysis of sound anomaly detection techniques: Existing methods and our approach on three published datasets.

Dataset	Study	Model	Feature Extraction	ML	DL	SL/Se-SL/UL
MIMII DG	[72]	Transformer Decoder	Feature by PANNs	✗	✓	UL
	[73]	MobileNetV2	Mel-Spectrogram	✗	✓	UL
	[74]	MobileNetV2, GMM	Log Mel-Spectrogram	✓	✓	UL
	[75]	MPE, ResNet18	Log Mel-Spectrogram	✗	✓	UL
	[76]	FFT, Sincnet, Log-mel	Log Mel-Spectrogram	✗	✓	UL
	Ours	SVM, DT, XGBoost RNN, LSTM, GRU	Mel-Spectrogram, MFCCs	✓	✓	SL
DCASE 2022	[64]	AutoEncoder, MobileNetV2	✗	✗	✓	UL
	[77]	EfficientNetV2	Log Mel-Spectrogram	✗	✓	UL
	[78]	MobileNetV2 baseline	Spectrogram by STFT	✗	✓	UL
	[79]	GMM-based clustering	Spectral-temporal fusion	✓	✗	UL
	[80]	Tgram + Pre-trained	Log Mel-Spectrogram	✗	✓	UL
	[81]	Domain Generalization, Domain Specialization	Mel-Spectrogram	✗	✓	UL
	[82]	DP-MAE	Log Mel-Spectrogram	✗	✓	Se-SL
	[83]	SSL	Spectrogram	✗	✓	Se-SL
	Ours	SVM, DT, XGBoost RNN, LSTM, GRU	Mel-Spectrogram, MFCCs	✓	✓	SL
DCASE 2024	[65]	Autoencoder	STFT	✗	✓	UL
	[84]	AE	Mel-spectrogram	✗	✓	UL-SL
	Ours	SVM, DT, XGBoost RNN, LSTM, GRU	Mel-Spectrogram, MFCCs	✓	✓	SL

Acronyms: SL: Supervised Learning; Se-SL: Semi-Supervised Learning; UL: Unsupervised Learning;
Legends: ✓ means included, ✗ means not included.

machinery, achieving high accuracy in fault diagnosis and condition monitoring. Similarly, research by [19], [20], [21], [22], and [23] utilized Decision Tree (DT) or Forest models to classify abnormal sound patterns in industrial environments, enabling the early detection of equipment malfunctions and failures. Furthermore, the XGBoost model has also been applied to detect anomalies [22], [24], [25], [26], [27], [28].

In recent years, DL techniques, particularly Recurrent Neural Networks (RNNs), and their variants, have emerged as powerful tools for anomaly detection in sequential data, such as time-series signals and audio recordings. RNNs, equipped with memory cells that retain information over time, have shown promising results in capturing temporal dependencies and detecting subtle deviations from normal patterns. For instance, studies by [18], [29], [30], [31], [32], [33] applied Long Short-Term Memory (LSTM) networks to analyze acoustic signals for fault detection in industrial machinery, achieving superior performance compared to traditional ML methods. In addition, Gated Recurrent Units (GRUs), a simplified version of LSTM, have been utilized for anomaly detection in audio data [40], [42], [43], [44].

Researchers have extensively investigated various feature extraction methods to improve anomaly detection accuracy in industrial sound analysis. Among these methods, Mel-Spectrogram and Mel-frequency cepstral coefficients

(MFCCs) have been widely utilized to transform raw audio signals into informative representations capturing relevant acoustic characteristics. Mel-Spectrogram provides a visual representation of the frequency content of sound signals over time and have been applied in diverse industrial applications, such as fault diagnosis in rotating machinery and leak detection in pipelines [45], [46], [47], [48], [49], [50], [51], [52]. Similarly, MFCCs, which extract spectral characteristics based on the response of the human auditory system, have been effective in detecting abnormal sound patterns in automotive engines and manufacturing processes [53], [54], [55], [56], [57], [58], [59], [60], [61].

In the existing literature, researchers have proposed various techniques for the detection of sound anomalies in industrial settings, each utilizing different models and feature extraction methods. For example, Kawaguchi et al. [71] used a transformer detector model with features extracted by PANNs, achieving promising results in domain generalization tasks. Abdelhady et al. [72] utilized MobileNetV2 with Mel-Spectrogram features for anomaly detection, demonstrating the effectiveness of detecting abnormal sound patterns. Similarly, Amer et al. [73] combined MobileNetV2 with Gaussian Mixture Models (GMM) on log Mel-Spectrogram features, achieving notable results in detecting anomalies in industrial sound recordings.

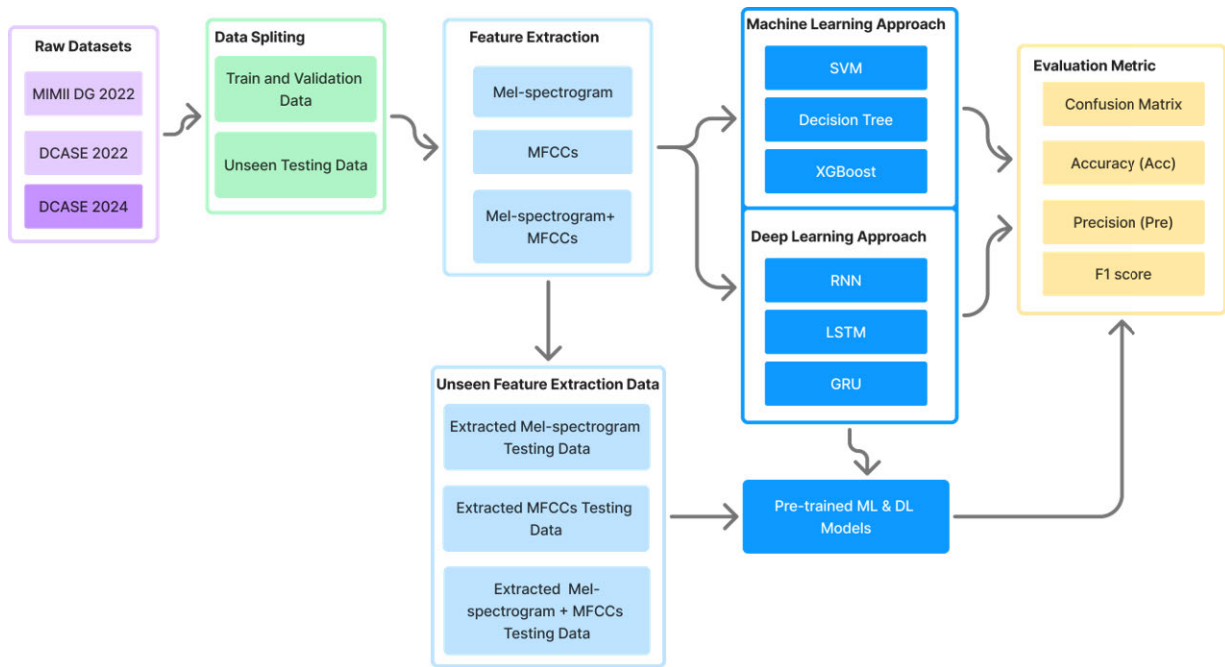


FIGURE 1. The proposed anomaly industrial machine sound detection.

While existing studies have significantly contributed to the field, notable gaps and limitations in prior methods remain. Many approaches focus solely on either ML or DL techniques, overlooking the potential benefits of a hybrid approach. Additionally, the reliance on Mel-Spectrogram features for sound representation neglects the complementary information provided by other feature extraction methods like MFCCs. Furthermore, the lack of comprehensive evaluation across different datasets and model architectures limits the generalizability and robustness of existing anomaly detection techniques.

To address these gaps, our proposed approach integrates both ML and DL techniques, including SVM, DT, XGBoost, RNN, LSTM, and GRU models, to leverage the strengths of each paradigm. We also explore the efficacy of both Mel-Spectrogram and MFCCs features for sound representation, aiming to capture a broader range of acoustic characteristics inherent in industrial machinery. By conducting extensive evaluations on three distinct datasets, the MIMII DG [62], DCASE 2022 [63], and DCASE 2024 [64] datasets, we aim to provide a comprehensive analysis of anomaly detection techniques in real-world industrial environments, highlighting the advantages of our approach over existing methods. Our comparative analysis, as shown in Table 1, demonstrates how our proposed approach fills the gaps left by prior methods, offering superior performance and greater adaptability in detecting anomalous sound patterns across diverse industrial contexts.

In contrast to previous studies, our work presents several novel contributions to the field of industrial sound anomaly detection. Firstly, while many existing approaches rely on traditional ML or DL techniques in isolation, we propose

a hybrid framework that integrates both ML (SVM, DT, XGBoost) and DL models (RNN, LSTM, GRU), combining the strengths of each paradigm to improve detection accuracy and robustness. Additionally, unlike previous methods that predominantly rely on Mel-Spectrogram features, we explore using both Mel-Spectrogram and MFCCs, capturing a wider range of acoustic characteristics for enhanced anomaly detection. Furthermore, our work introduces a comprehensive evaluation across two distinct datasets, MIMII DG and DCASE 2022, offering a broader and more rigorous analysis of model performance in real-world industrial environments. This approach not only addresses the limitations in generalizability observed in previous studies but also provides valuable insights into the adaptability of our hybrid model to various industrial applications. By filling these critical gaps, our research significantly advances current state-of-the-art industrial anomaly detection.

III. METHODOLOGY

A. OVERVIEW OF THE APPROACH

The methodology adopted in this study revolves around utilizing a combination of ML and DL techniques for anomalous sound detection in industrial environments, as depicted in Figure 1. The approach involves feature extraction from the raw sound data, model training using both ML and DL algorithms, and performance evaluation to assess the effectiveness of the proposed method.

B. DATASET PREPARATION

The MIMII DG and DCASE 2022 datasets are crucial components of this study. They provide diverse and com-

prehensive collections of sound recordings captured in real-world industrial environments.

- MIMII DG dataset [62]. The MIMII DG dataset is a valuable resource for investigating malfunctioning industrial machines and conducting domain generalization tasks. It comprises normal and abnormal operating sounds from various industrial machines, including fans, gearboxes, bearings, slide rails, and valves. Each machine type is subdivided into three sections, introducing different domain shifts to simulate real-world scenarios. This dataset offers a diverse range of sound recordings capturing the acoustic signatures of machinery operating under different conditions, facilitating the study of anomaly detection algorithms.
- DCASE 2022 dataset [63]. In contrast, the DCASE 2022 dataset provides a broader spectrum of industrial sound environments, featuring recordings from manufacturing plants, transportation systems, construction sites, and other industrial settings. It includes seven types of real/toy machines: fans, gearboxes, bearings, slide rails, toycars, toytrains, and valves. Each recording is a single-channel and 10-sec-long audio with a sampling rate of 16 kHz. The dataset consists of multiple sections for each machine type, offering a comprehensive training set and test data for evaluating anomaly detection algorithms. Each recording includes the machine's operating sound and environmental noise, presenting challenges typical of real-world industrial environments.
- DCASE 2024 dataset [64]. The dataset comprises normal and anomalous operational sounds from nine different types of real and toy machines. Each recording is a single-channel audio clip capturing both the machine's operating sound and background environmental noise. The duration of these recordings ranges between 6 and 10 seconds. The machines included in this dataset are: 3D Printer, Air Compressor, Brushless Motor, Hair Dryer, Hovering Drone, Robotic Arm, Scanner, Toothbrush, and Toy Circuit.

Table 2 provides a detailed description of the training and testing data from three datasets, including the number of sound wave files, the presence of normal and anomalous sounds, and attributes related to domain shifts. These datasets serve as essential resources for benchmarking anomaly detection algorithms and advancing research in industrial sound analysis.

Three datasets undergo preprocessing before model training to ensure data quality and consistency. This preprocessing step involves removing background noise, normalizing sound levels, and segmenting recordings into uniform time frames. By standardizing the duration of sound segments, the datasets facilitate feature extraction and model training, enabling a systematic and consistent analysis of sound data across different recordings and industrial contexts.

Overall, this study's utilization of the MIMII DG dataset, the DCASE 2022 dataset, and the DCSE 2024 dataset

enhances the robustness and generalizability of the proposed approach for anomalous sound detection in industrial environments. These datasets provide valuable insights into the acoustic characteristics of industrial machinery and enable comprehensive evaluations of anomaly detection algorithms under diverse real-world conditions.

C. FEATURE EXTRACTION

High-frequency features are extracted from raw sound recordings to capture the distinctive characteristics of industrial sound data. Two commonly used techniques for feature representation in sound processing are Mel-spectrogram and Mel-frequency spectral coefficients (MFCCs).

High-frequency features are extracted from raw sound recordings to capture the distinctive characteristics of industrial sound data. Specifically, Mel-Spectrogram and Mel-Frequency Cepstral Coefficients (MFCCs) are employed as feature representations. Mel-Spectrogram provide a time-frequency representation of sound signals, while MFCCs capture the spectral features of the sound. These features are extracted using standard signal processing techniques such as the Short-Time Fourier Transform (STFT) and Mel-frequency scaling.

1) MEL-SPECTROGRAM BASED ON FAST FOURIER TRANSFORM (FFT)

The Mel-Spectrogram is a widely used representation in audio signal processing, particularly for tasks such as speech recognition and sound classification. It provides a time-frequency representation of an audio signal, where the frequency axis is warped according to the Mel scale to better align with human auditory perception.

Fast Fourier Transform (FFT): The computation of the Mel-Spectrogram begins with applying FFT to the audio signal. The FFT decomposes the signal into its frequency components, providing a spectrum representation of the signal at each time frame.

The FFT algorithm can be described recursively as follows: Divide: Split the sequence $x[n]$ into even-indexed and odd-indexed subsequences. Conquer: Compute the Discrete Fourier Transform (DFT) of the even-indexed and odd-indexed subsequences recursively. Combine: Combine the results of the even and odd DFTs to obtain the final DFT. Mathematically, this can be expressed as:

$$X[k] = X_e[k] + e^{-j2\pi k/N} \cdot X_o[k], \quad (1)$$

where $X_e[k]$ and $X_o[k]$ are the DFTs of the even and odd subsequences, respectively.

Mel-Frequency Warping: After the FFT is computed, the frequency axis is transformed into the Mel-frequency scale. The Mel scale is a perceptually-based scale that approximates the human auditory system's response to sound frequencies. The conversion from Hertz (Hz) to Mel scale is typically

TABLE 2. Description of training and testing data from three datasets.

Dataset	Data Part	Data Types	Device	No. Recording Files	Duration/Each File (s)
MIMII DG	Training	Normal	Bearing	2,999	10
			Gearbox	3,000	10
			Fan	3,000	10
			Slider	3,000	10
			Valve	3,000	10
	Testing	Normal & Anomaly	Bearing	600	10
			Gearbox	600	10
			Fan	600	10
			Slider	600	10
			Valve	600	10
DCASE 2022	Training	Normal	Bearing	2,999	10
			Gearbox	3,000	10
			Fan	3,000	10
			Slider	3,000	10
			Valve	3,000	10
			ToyTrain	3,000	10
			ToyCar	3,000	10
	Testing	Normal & Anomaly	Bearing	600	10
			Gearbox	600	10
			Fan	600	10
			Slider	600	10
			Valve	600	10
			ToyTrain	600	10
DCASE 2024	Training	Normal	3DPrinter	1,000	[6-10]
			AirCompressor	1,000	[6-10]
			BrushlessMotor	1,000	[6-10]
			HairDryer	1,000	[6-10]
			HoveringDrone	1,000	[6-10]
			RoboticArm	1,000	[6-10]
			Scanner	1,000	[6-10]
			ToothBrush	1,000	[6-10]
			ToyCircuit	1,000	[6-10]
	Testing	Normal & Anomaly	AirCompressor	200	[6-10]
			BrushlessMotor	200	[6-10]
			HairDryer	200	[6-10]
			HoveringDrone	200	[6-10]
			RoboticArm	200	[6-10]
			Scanner	200	[6-10]
			ToothBrush	200	[6-10]
			ToyCircuit	200	[6-10]

achieved using the following formula:

$$M(f) = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right), \quad (2)$$

where $M(f)$ is the frequency in the Mel scale; f is the frequency in Hertz; 2595 is a constant factor utilized in the Mel-frequency scaling formula to convert frequency

values from Hertz to the Mel scale. This value is derived from the formula designed to convert frequencies to the Mel scale, considering the characteristics of human auditory perception. Additionally, 700 serves as a constant factor in the Mel-frequency scaling formula. It represents a reference frequency (in Hertz) that aligns with the lower edge of the Mel scale. This value is empirically determined based on psycho-acoustic studies to approximate the non-linear frequency perception of human hearing.

Mel Filter Bank: After converting the frequency axis to the Mel scale, a set of triangular filters, known as the Mel filter bank, is applied to the spectrum obtained from the FFT. Each filter in the bank corresponds to a specific Mel-frequency bin. The energy within each filter bank is then summed to obtain the Mel-scaled spectrogram.

Mel-Scaled Spectrogram: The Mel-scaled spectrogram is obtained by multiplying the magnitude of the FFT spectrum by the values of the Mel filter bank. Mathematically, the Mel-scaled spectrogram $S_m(t, m)$ at time t and Mel-frequency bin m is given by:

$$S_m(t, m) = \sum_{\omega} |X(t, \omega)|^2 \cdot H_m(\omega), \quad (3)$$

where $S_m(t, m)$ is the Mel-scaled spectrogram at time t and Mel-frequency bin m ; $X(t, \omega)$ is the magnitude of the FFT spectrum at time t and frequency ω ; $H_m(\omega)$ is the value of the Mel filter bank for the Mel-frequency bin m at frequency ω .

The Mel-Spectrogram provides a compact representation of the frequency content of the audio signal, with the frequency axis warped according to the Mel scale to better align with human perception.

2) MEL-FREQUENCY CEPSTRAL COEFFICIENTS (MFCCS) BASED ON FFT AND DISCRETE COSINE TRANSFORM (DCT)

Mel-Frequency Cepstral Coefficients (MFCCs) are widely used feature representations in speech and audio signal processing. They are particularly effective for speech recognition, speaker identification, and sound classification. MFCCs are derived from the Mel-Spectrogram, computed using the Fast Fourier Transform (FFT) and subsequent Mel-frequency scaling. After obtaining the Mel-Spectrogram, the MFCCs are computed using the Discrete Cosine Transform (DCT).

Mel-Spectrogram Calculation (FFT and Mel-Frequency Warping):

- 1) **FFT:** The audio signal is divided into short overlapping frames. The FFT is applied to each frame to obtain the frequency spectrum.
- 2) **Mel-Frequency Warping:** The frequency axis is warped into the Mel scale using the Mel-frequency scaling formula 2. This converts the linear frequency scale (Hertz) into the Mel scale, better aligning with human auditory perception.

Logarithmic Compression: After filtering the spectrum, the logarithm of the filter-bank energies is taken. This step is crucial for mimicking the non-linear nature of human

auditory perception. Mathematically, the logarithm of the filter-bank energies is calculated as:

$$E_m = \log \left(\sum_k |X_k|^2 H_m(k) \right), \quad (4)$$

where E_m is the log filter-bank energy for the $m - th$ Mel-filter; X_k is the STFT coefficient at frequency index k ; $H_m(k)$ is the filter-bank response for the $m - th$ Mel-filter at frequency index k .

Discrete Cosine Transform (DCT): Finally, the Discrete Cosine Transform (DCT) is applied to the log filter-bank energies to decorrelate the features and extract a set of coefficients. The DCT formula is given by:

$$C_n = \sum_{m=0}^{M-1} E_m \cdot \cos \left[\frac{\pi}{M} \left(m + \frac{1}{2} \right) n \right], \quad (5)$$

where C_n is the $n - th$ MFCC coefficient; E_m is the log filter-bank energy for the $m - th$ Mel-filter; M is the number of Mel-filter banks.

The resulting C_n coefficients are MFCCs, which provide a compact representation of the signal's spectral characteristics. These coefficients are commonly used as features for various audio signal processing tasks, such as speech recognition, speaker identification, and sound classification.

MFCCs provide a compact and effective representation of the spectral envelope of an audio signal. By capturing the characteristics of the human auditory system, they are well-suited for various speech and audio processing tasks. Combining FFT for spectral analysis, Mel-frequency scaling for perceptual frequency warping, and DCT for decorrelation results in a powerful feature representation for speech and audio signal processing applications.

D. MODEL TRAINING

In this study, both ML and DL algorithms are employed for sound detection, each offering unique capabilities for capturing and analyzing the complex patterns present in industrial sound data.

Various classifiers are utilized for ML algorithms, including Support Vector Machine (SVM), Decision Tree (DT), and XGBoost. These classifiers are trained on labeled data, where each sound recording is labeled as a normal or anomalous event. During training, the classifiers learn to identify patterns and characteristics that distinguish between normal operational sounds and anomalous sound events. By leveraging features extracted from the sound data, such as Mel-Spectrogram or MFCCs, the ML algorithms can effectively classify new sound samples and detect anomalies in industrial machinery sound data.

- **SVM (Support Vector Machine)** is a supervised learning algorithm for classification tasks. It works by finding the optimal hyperplane that best separates the data into different classes [65]. Given a set of training data (X, y) , where X represents the input features (e.g., Mel-Spectrogram or MFCCs) and y represents the

corresponding labels (normal or anomaly), SVM aims to find the hyperplane $w_T x + b = 0$ that maximizes the margin between the classes, subject to the constraint $y_i(w_T x_i + b) \geq 1$. The SVM model is trained using the input features extracted from the high-frequency data. During training, the SVM algorithm optimizes the hyperplane parameters w and b to achieve maximum class margin separation.

- DT (Decision Tree) is a tree-like model for classification tasks. It partitions the data recursively based on the feature values to create a tree structure that predicts the class labels [66]. DT recursively splits the feature space into regions to minimize each node's impurity. The splitting criterion, such as Gini impurity or entropy, determines the best feature and threshold for splitting. Let D be a dataset of n samples, each represented by a feature vector x_i of dimension m and corresponding class labels y_i . The goal of DT is to partition the feature space recursively into regions where each region corresponds to a decision node in the tree. DT selects the best feature j and threshold θ at each tree node to split the data into two subsets D_L and D_R . This splitting is determined based on a criterion such as Gini impurity or entropy, which measures the impurity of the data at each node. For a binary classification problem, let p_{jk} denote the proportion of samples in subset D_j that belong to class k . Gini impurity is defined as:

$$Gini(D) = 1 - \sum_{k=1}^K (p_{jk})^2 \quad (6)$$

Entropy impurity is defined as:

$$Entropy(D) = - \sum_{k=1}^K p_{jk} \log(p_{jk}) \quad (7)$$

DT selects the feature j and threshold θ that minimize the impurity after the split, i.e.,

$$Gain(D, j, \theta) = Impurity(D) - \left(\frac{|D_L|}{|D|} Impurity(D_L) + \frac{|D_R|}{|D|} Impurity(D_R) \right) \quad (8)$$

where the feature j and threshold θ that maximizes the information gain are chosen for splitting. The tree-building process continues recursively until a stopping criterion is met, such as reaching a maximum depth, minimum number of samples per leaf node, or minimum impurity threshold. Once the tree is constructed, a prediction is made by traversing the tree based on the feature values of the input sample until a leaf node is reached. The majority class in the leaf node is then assigned as the predicted class label for the input sample. DT model is trained using the input features extracted from the high-frequency data. During training, the Decision Tree algorithm recursively partitions the

feature space based on the feature values to create a tree structure that predicts the class labels.

- XGBoost (Extreme Gradient Boosting) is an ensemble learning algorithm that combines the predictions of multiple weak learners (decision trees) to improve overall prediction accuracy [67]. XGBoost optimizes a differentiable loss function by iterative adding decision trees to the ensemble. Each tree is trained to correct the errors made by the previous trees. XGBoost aims to build an ensemble of weak learners (decision trees) and combine their predictions to form a strong predictive model. XGBoost optimizes an objective function consisting of the loss function and a regularization term. The loss function measures the discrepancy between the predicted scores ($\hat{y}$) and the true labels (y). For binary classification, a common choice is the logistic loss:

$$loss(y, \hat{y}) = \sum_{i=1}^n \left[y_i \log(1 + e^{-\hat{y}_i}) + (1 - y_i) \log(1 + e^{\hat{y}_i}) \right] \quad (9)$$

The regularization term penalizes the complexity of the model to prevent overfitting. Common choices include $L1$ and $L2$ regularization. XGBoost employs a gradient-boosting algorithm to add decision trees to the ensemble iteratively. At each iteration, a new decision tree fits the negative gradient of the loss function concerning the predictions of the current ensemble. Each decision tree is grown using a greedy algorithm that recursively splits the data into regions based on the feature values. The splitting criterion is determined by optimizing the objective function, considering both the improvement in loss and the regularization term. XGBoost applies regularization techniques such as tree pruning to control the complexity of individual trees and prevent overfitting. Trees are pruned based on their depth or the number of leaf nodes. The predictions of individual trees are combined using a weighted sum to obtain the final ensemble prediction. The weights of the trees are determined during the training process based on their contribution to reducing the loss function. The XGBoost model is trained using the input features extracted from the high-frequency data. During training, XGBoost iteratively adds decision trees to the ensemble, optimizing the loss function to improve prediction accuracy.

In contrast, the DL approach focuses on leveraging the power of recurrent neural network (RNN) architectures to capture temporal dependencies in the sound data. Specifically, LSTM and Gated Recurrent Unit (GRU) networks are employed because they can model sequential data and capture long-term dependencies effectively. These RNN architectures are trained on sequential sound data, where each input represents a time series of sound features extracted from consecutive time frames. By learning the temporal dynamics of sound patterns, LSTM and GRU networks can detect

anomalies in time-series data, making them well-suited for analyzing industrial sound recordings where anomalies may manifest over time.

- RNN (Recurrent Neural Network): RNNs are designed to process sequential data by maintaining an internal state or memory that captures information from previous time steps [68]. Given an input sequence $x = (x_1, x_2, \dots, x_T)$, where x_t represents the input at time step t , and an initial hidden state h_0 , the hidden state h_t at time step t is computed as:

$$h_t = f(x_t, h_{t-1}), \quad (10)$$

where f is a non-linear activation function (typically a sigmoid or hyperbolic tangent function) that combines the input x_t and the previous hidden state h_{t-1} to compute the current hidden state h_t . The output at each time step can be computed using the hidden state h_t and a weight matrix W :

$$y_t = \text{softmax}(Wh_t) \quad (11)$$

RNNs suffer from the vanishing gradient problem, where gradients diminish exponentially over long sequences, making it difficult to learn long-range dependencies.

- LSTM (Long Short-Term Memory): LSTM networks address the vanishing gradient problem by introducing a gating mechanism that regulates the flow of information through the network [69]. At each time step, an LSTM unit maintains a cell state C_t and three gates: input gate i_t , forget gate f_t , and output gate o_t . The input gate controls the flow of new information into the cell state, while the forget gate determines which information to discard from the cell state, and the output gate regulates the information to be output from the cell state. The computations within an LSTM unit can be expressed as follows:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}C_{t-1} + b_i) \quad (12)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}C_{t-1} + b_f) \quad (13)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}C_{t-1} + b_o) \quad (14)$$

$$g_t = \tanh(W_{xg}x_t + W_{hg}h_{t-1} + b_g) \quad (15)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot g_t \quad (16)$$

$$h_t = o_t \odot \tanh(C_t) \quad (17)$$

where σ represents the sigmoid activation function, $\odot$ denotes element-wise multiplication, and W and b are weight matrices and bias vectors, respectively.

- GRU (Gated Recurrent Unit): GRU is a variation of LSTM that simplifies the architecture by combining the forget and input gates into a single update gate and merging the cell state and hidden state [70]. Similar to LSTM, GRU also maintains a hidden state h_t and a cell state C_t , but it computes these states differently:

$$z_t = \sigma(W_{xz}x_t + W_{hz}h_{t-1} + b_z) \quad (18)$$

$$r_t = \sigma(W_{xr}x_t + W_{hr}h_{t-1} + b_r) \quad (19)$$

TABLE 3. General architecture of RNN, LSTM, or GRU model in our approach.

Layer Name	Layer Structure	Output Size
Bidirectional RNN/LSTM/GRU	32	13×64
Dropout	0.1	13×64
Bidirectional RNN/LSTM/GRU	32	13×64
Dropout	0.1	13×64
Bidirectional RNN/LSTM/GRU	64	128
Dropout	0.2	128
Dense	64	64
Dropout	0.1	64
Dense	64	64
Dropout	0.1	64
Dense	64	64
Dropout	0.2	64
Dense	128	128
Dropout	0.2	128

$$\hat{h}_t = \tanh(W_{xh}x_t + W_{hh}(r_t \odot h_{t-1}) + b_h) \quad (20)$$

Our approach employs a sophisticated architecture comprising RNN, LSTM, and GRU models, as depicted in Table 3. By leveraging this architecture, our approach aims to harness the strengths of RNN, LSTM, and GRU models for anomaly detection tasks, ensuring robust performance across diverse industrial sound datasets.

This architecture is meticulously crafted to effectively capture temporal dependencies and subtle deviations in the input data. The input layer accommodates various sizes depending on the feature input: for instance, Mel-Spectrogram with an input shape of 128×313 , MFCCs with an input shape of 12×313 , or a combination of both features with an input shape of 128×313 . The Bidirectional RNN/LSTM/GRU layers as Feature Extractor, sized at 32, play a pivotal role in processing the input data bidirectionally, enhancing the model's capacity to assimilate information from past and future contexts. Dropout layers with a dropout rate of 0.1 are strategically placed after each Feature Extractor layer to mitigate overfitting by randomly dropping out a fraction of input units during training. Subsequent Feature Extractor layers refine the input data representation, augmenting the model's ability to discern intricate patterns. To ensure regularization, an additional set of Dropout layers with a slightly elevated dropout rate of 0.2 follows these layers. The Dense layers, each configured with a size of 64, serve as the neural network's backbone, transforming the input data nonlinearly to extract higher-level features. Dropout layers with a dropout rate of 0.1 are interspersed between the Dense layers to prevent the model from excessively relying on specific features and to enhance generalization. Ultimately, the Dense Output layer, whose size varies based on the number of classes in the classification task, produces the model's final output. This work has five and seven Dense outputs corresponding to MIMII DG and DCASE 2022 datasets. This layer applies an appropriate activation function, such as softmax for multi-class classification, to generate class probabilities.

ML and DL models are optimized using labeled training data to minimize classification errors and improve overall accuracy during training. The models undergo iterative training, where parameters are adjusted based on the observed performance of the training data. Additionally, cross-validation and hyperparameter tuning may be employed to optimize the models' performance further and prevent overfitting to the training data.

Overall, using ML and DL algorithms for sound detection enables a comprehensive approach to anomaly detection in industrial environments. By combining the strengths of traditional ML classifiers with the advanced capabilities of DL architectures, the proposed methodology offers a robust and effective solution for detecting anomalous sound events and ensuring operational safety and efficiency in industrial settings.

E. PERFORMANCE EVALUATION

Performance evaluation uses standard metrics such as confusion matrix, accuracy, precision, recall, and F1-score. The trained models are evaluated on separate test datasets to assess their ability to classify normal and anomalous sound events accurately.

- Confusion Matrix:

$$CM = \begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix} \quad (21)$$

where TN (True Negative): Number of correctly classified normal sound events. FP (False Positive): Number of normal sound events incorrectly classified as anomalies. FN (False Negative): Number of anomalous sound events incorrectly classified as normal. TP (True Positive): Number of correctly classified anomalous sound events.

- Accuracy: Accuracy measures the proportion of correctly classified instances (both normal and anomalous) out of the total instances in the dataset. It is calculated using the following formula:

$$Accuracy = \frac{TN + TP}{TN + FP + FN + TP} \quad (22)$$

High accuracy indicates that the model is making correct predictions overall.

- Precision: Precision measures the proportion of correctly classified anomalous instances out of all instances predicted as anomalous by the model. It is calculated using the following formula:

$$Precision = \frac{TP}{TP + FP} \quad (23)$$

Precision quantifies the model's ability to avoid false alarms and correctly identify anomalous sound events.

- Recall: Recall, also known as sensitivity or true positive rate, measures the proportion of correctly classified anomalous instances out of all actual anomalous

instances in the dataset. It is calculated using the following formula:

$$Recall = \frac{TP}{TP + FN} \quad (24)$$

Recall evaluates the model's ability to capture all anomalous sound events without missing any.

- F1-Score: The F1-score is the harmonic mean of precision and recall. It provides a single metric that balances both precision and recall. It is calculated using the following formula:

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (25)$$

The F1-score ranges from 0 to 1, with higher values indicating a better balance between precision and recall.

The trained models are evaluated on separate test datasets to assess their performance. Each model's predictions are compared against ground truth labels to compute the confusion matrix. The confusion matrix calculates accuracy, precision, recall, and F1-score to quantify the model's ability to classify normal and anomalous sound events accurately.

IV. EXPERIMENTAL SETUP

A. EXPERIMENTAL DESIGN

The experimental design encompasses several key steps to ensure robustness and reproducibility when evaluating the proposed anomaly detection methods.

Firstly, data preprocessing is conducted on industrial sound datasets to prepare them for model training and evaluation. This involves labeling the output using a `LabelEncoder` to convert categorical labels into numerical representations, facilitating model interpretation and analysis. The `LabelEncoder` is a preprocessing step that converts categorical labels into numerical representations, facilitating model training and evaluation. Mathematically, the `LabelEncoder` assigns a unique integer to each distinct category present in the label data. Let y denote the original categorical labels, and let $LE(y)$ represent the numerical encoding obtained after applying the `LabelEncoder`. The encoding process can be described as follows: Given a set of N unique categories in the label data, indexed from 0 to $N-1$, the `LabelEncoder` assigns an integer label i to each category c such that $0 \leq i < N$.

$$LE(y) = \begin{cases} 0 & \text{if } y_i = \text{category}_1 \\ 1 & \text{if } y_i = \text{category}_2 \\ \vdots & \vdots \\ N-1 & \text{if } y_i = \text{category}_n, \end{cases} \quad (26)$$

where y_i represents the i_{th} element of the original categorical label vector y .

After applying the `LabelEncoder`, the original categorical labels are replaced with their corresponding numerical representations, enabling these labels in ML algorithms that require numerical inputs. This preprocessing step ensures

TABLE 4. Parameter setting for high-frequency feature extraction methods.

Feature Name Method	Parameter Name	Denote	Value
Mel-Spectrogram	No. of the FFT window	<code>n_fft</code>	128
	Sampling rate	<code>sr</code>	24.000
	No. of samples between successive frames	<code>hop_length</code>	512
	Window function	<code>window</code>	'hann'
	Frame is centered	<code>center</code>	True
	Padding mode to use at the edges of the signal	<code>pad_mode</code>	'constant'
	Exponent for the magnitude Mel-Spectrogram	<code>power</code>	2.0
MFCCs	Sampling rate	<code>sr</code>	24.000
	No. of MFCCs to return	<code>n_mfcc</code>	13
	Discrete cosine transform (DCT) type	<code>dct_type</code>	2
	Normalization	<code>norm</code>	'ortho'
	The higher-order coefficients	<code>lifter</code>	0

compatibility between the data and the models, facilitating accurate and efficient model training and evaluation.

Next, the preprocessed data is split into training and validating sets using an 80:20 ratio. This partitioning ensures that sufficient data is available for model training while preserving a separate portion for evaluation to assess generalization performance accurately.

In terms of implementation, the proposed methods are developed using the Python programming language and relevant libraries such as TensorFlow 2.10.0, scikit-learn 1.3.2, and librosa 0.10.1 for sound processing and ML tasks. These libraries offer extensive data manipulation, feature extraction, model training, and evaluation functionalities, streamlining the implementation process and enabling efficient experimentation.

Additionally, hardware requirements are considered to ensure the smooth execution of the experimental procedures. While specific hardware specifications may vary depending on the dataset size and computational complexity of the models, a standard computing environment with sufficient processing power and memory capacity is typically adequate for experimenting: Windows 10, CPU: i9 10900K, RAM: 64 GB, GPU: RTX 3080 Ti.

Following these steps in the experimental design, we aim to establish a systematic and rigorous framework for evaluating the proposed anomaly detection methods, enabling reliable comparisons and insights into their performance across different datasets and experimental conditions.

B. HYPER-PARAMETERS TUNING

For the feature extraction method, we carefully selected and tuned parameters for each method to optimize the extraction of Mel-Spectrogram and MFCCs features, as detailed in Table 4.

The parameters of the Mel-Spectrogram extraction method encompass several key aspects crucial for accurate feature representation. The number of FFT windows (`n_fft`)

determines the frequency resolution, influencing the level of detail captured in the spectrogram. Sampling rate (`sr`) defines the frequency range analyzed, ensuring that relevant spectral components are captured within the spectrogram. Hop length (`hop_length`) regulates the overlap between successive frames, balancing temporal resolution with computational efficiency. Window function (`window`) selection influences spectral leakage and sidelobe levels, impacting the accuracy of frequency representation. Frame centering (`center`) determines whether frames are centered, affecting temporal alignment within the spectrogram. Padding mode (`pad_mode`) addresses edge effects during analysis, enhancing the robustness of spectrogram computation. Finally, the magnitude exponent (`power`) adjusts the dynamic range of the spectrogram, optimizing the representation of signal magnitudes in the frequency domain. Collectively, these parameters allow fine-tuning of the Mel-Spectrogram extraction process to capture relevant spectral characteristics accurately.

The parameters involved in extracting MFCCs play a crucial role in effectively capturing the spectral characteristics of audio signals. The sampling rate (`sr`) determines the frequency range analyzed, ensuring that the appropriate spectral components are represented in the coefficients. The number of MFCCs to return (`n_mfcc`) controls the dimensionality of the feature space, balancing between capturing relevant spectral information and computational efficiency. The choice of discrete cosine transform (DCT) type (`dct_type`) influences the coefficients' decorrelation, affecting the feature set's discrimination power. Normalization (`norm`) adjusts the magnitude of the coefficients to ensure consistency across different signals, enhancing the robustness of the feature extraction process. Additionally, the lifter parameter determines the emphasis on higher-order coefficients, impacting the emphasis on specific spectral components in the feature representation. Together, these parameters allow for the customization of the MFCC

TABLE 5. Hyper-parameter setting for our ML and DL models.

Model Name	Hyperparameters Name	Value
SVM	Specifies the kernel type to be used (<i>kernel</i>)	'linear'
	Regularization parameter (<i>C</i>)	1
	Degree of the polynomial kernel function (<i>degree</i>)	3
	Kernel coefficient (<i>gamma</i>)	'scale'
	Tolerance for stopping criterion (<i>tol</i>)	0.001
	Specify the size of the kernel cache (<i>cache_size</i>)	200
	Hard limit on iterations within solver (<i>max_iter</i>)	-1
	Decision function of shape (<i>decision_function_shape</i>)	'ovr'
DT	Function to measure the quality of a split (<i>criterion</i>)	'gini'
	Strategy used to choose the split (<i>splitter</i>)	'best'
	Minimum no. of samples to split (<i>min_samples_split</i>)	2
	Minimum no. of samples required to be at a leaf node (<i>min_samples_leaf</i>)	1
XGBoost	Number of estimator (<i>n_estimators</i>)	2
	Max depth (<i>max_depth</i>)	2
	Learning rate (<i>learning_rate</i>)	1
	Loss function (<i>objective</i>)	'multi:softprob'
RNN, LSTM, GRU	Learning rate	0.0001
	Batch size	32
	Dropout	0.1
	Optimizer	Adam
	Number of epochs	30
	Activation	'softmax'
	Loss function	'CategoricalCrossentropy'

extraction process to capture relevant spectral features accurately for subsequent analysis and classification tasks.

The hyperparameters for ML and DL models are carefully tuned to optimize their performance in anomaly detection tasks, as shown in Table 5.

For ML models like SVM and DT, parameters such as the kernel type, regularization parameter, and criteria for split quality are adjusted to enhance model accuracy. In SVM, the choice of kernel type, regularization parameter (*C*), and kernel coefficient (*gamma*) are crucial for defining decision boundaries and controlling overfitting. Similarly, the split criteria, minimum samples per split, and minimum impurity decrease determine DT's tree structure and decision-making process. XGBoost, a gradient boosting algorithm, is tuned by adjusting parameters like the number of estimators, maximum depth, and learning rate to optimize ensemble learning and improve predictive accuracy.

For DL models like RNN, LSTM, and GRU, hyperparameters such as learning rate, batch size, dropout rate, optimizer, number of epochs, activation function, and loss function are fine-tuned to improve model convergence and prevent overfitting. The learning rate controls the step size during optimization, while the batch size determines the

number of samples processed in each iteration. Dropout regularization is applied to prevent co-adaptation of feature detectors, enhancing model generalization. The choice of optimizers, such as Adam, influences the update strategy for model parameters. The number of epochs defines the number of iterations over the entire dataset during training, while the activation function and loss function determine the output format and optimization objective, respectively. By systematically adjusting these hyperparameters, the ML and DL models can effectively learn from the data and achieve high performance in anomaly detection tasks.

V. EXPERIMENTAL RESULTS AND DISCUSSION

The experimental results demonstrate the effectiveness of the proposed approach in detecting anomalous sounds within industrial environments. Performance metrics, including accuracy, precision, recall, and F1-score, are reported for both ML and DL models trained on three datasets: MIMII DG 2022, DCASE 2022, and DCASE 2024. This section presents two sets of results: the first focuses on sound anomaly detection, while the second examines multi-class classification. Additionally, the performance of the proposed

TABLE 6. Detection result of the proposed approach on MIMII DG 2022 dataset.

Model	Feature	Training Evaluation					Testing Evaluation				
		Acc	Pre	Rec	F1 Time (s)	Training	Acc	Pre	Rec	F1 Time (s)	Testing
SVM	MS	0.4880	0.5552	0.4872	0.4237	1944.7439	0.4636	0.5213	0.4636	0.4012	575.34
	MFCC	0.9498	0.9516	0.951	0.9510	204.408	0.8995	0.9009	0.8995	0.8985	127.3301
	Both	0.9498	0.9516	0.951	0.951	5629.1433	0.8995	0.9009	0.8995	0.8985	456.3277
DT	MS	0.7593	0.7612	0.7624	0.7618	676.2295	0.6871	0.7013	0.6871	0.6921	0.1373
	MFCC	0.9013	0.9036	0.9032	0.9033	372.0203	0.8079	0.8159	0.8079	0.8081	0.0195
	Both	0.9023	0.9041	0.9043	0.9042	1651.8284	0.8026	0.8146	0.8026	0.8049	0.1966
XGBoost	MS	0.961	0.9706	0.9707	0.9706	2084.0766	0.886	0.8926	0.886	0.8873	0.0984
	MFCC	0.9828	0.9832	0.9832	0.9832	814.7638	0.9102	0.9145	0.9002	0.8998	0.1511
	Both	0.9841	0.9844	0.9844	0.9844	2018.2837	0.9088	0.913	0.9088	0.9086	0.1478
RNN	MS	0.866	0.8744	0.8678	0.8679	8552.4948	0.796	0.8048	0.796	0.7928	23.491
	MFCC	0.9723	0.972	0.9728	0.9728	1063.2442	0.8997	0.9024	0.8996	0.898	3.1561
	Both	0.9583	0.9589	0.959	0.9589	12073.8299	0.8747	0.8813	0.8747	0.8731	24.7325
LSTM	MS	0.9317	0.9322	0.9324	0.9322	377.674	0.8823	0.8819	0.8823	0.8807	3.3453
	MFCC	0.9443	0.9461	0.945	0.9452	218.5663	0.869	0.8679	0.869	0.8666	1.782
	Both	0.958	0.9586	0.9586	0.9585	495.7487	0.9103	0.9119	0.9103	0.9097	4.8238
GRU	MS	0.9227	0.9255	0.9238	0.924	332.6181	0.8527	0.8677	0.8526	0.8529	2.7408
	MFCC	0.9683	0.9688	0.9688	0.9688	160.0866	0.922	0.9226	0.922	0.9206	1.0496
	Both	0.9543	0.9551	0.955	0.9549	373.1332	0.8987	0.9028	0.8987	0.8978	3.2442

method is compared with prior approaches. Finally, the section points out the advantage and limitation of our method.

A. PERFORMANCE EVALUATION OF THE PROPOSED APPROACH

We summarize each model's performance during two datasets' training and testing phases in Tables 6 (MIMII DG 2022) and 7 (DCASE 2022). The tables include performance metrics such as Accuracy (Acc), Precision (Pre), Recall (Rec), and F1 score. Acronyms are used, with MS denoting Mel-Spectrogram.

Based on the comprehensive analysis of the experimental results obtained from our methods, we can draw the following conclusion points:

- **ML Approach:** XGBoost consistently outperforms other ML models across datasets and feature sets, achieving remarkable accuracies exceeding 98% in the training phase and over 90% in the testing phase. Decision tree models demonstrate competitive performance, particularly achieving accuracies exceeding 90% in both training and testing phases.
- **DL Approach:** LSTM and GRU models exhibit strong performance in the DL approach, consistently achieving accuracies exceeding 95% across datasets and feature sets. While slightly trailing behind LSTM, GRU still demonstrates robust performance, particularly in the testing phase, where it achieves over 91% accuracy.
- **Feature Importance:** Both Mel-Spectrogram and MFCC features contribute significantly to the effectiveness of anomaly detection in industrial sound datasets. While Mel-Spectrogram features provide valuable insights, MFCC features offer additional discriminative power, particularly evident in the superior performance of the GRU model trained solely with MFCCs.
- **Overall Performance:** The hybrid approach, which combines supervised ML and DL techniques, showcases remarkable efficacy in anomalous sound detection in industrial environments. In the ML models, XGBoost with 2 features (Mel-Spectrogram and MFCCs) outperforms SVM and Decision Tree (DT) models. The combination of high-frequency features (Mel-Spectrogram and MFCCs) enhances the performance of the ML models in detecting anomalous sound events. In the DL approach, the GRU model, utilizing only MFCCs as a feature, demonstrates superior performance compared to the RNN and LSTM models. MFCCs alone provide sufficient information for the DL models to detect anomalous sound events effectively. In our analysis, the GRU outperformed the LSTM model in anomaly sound detection. Despite LSTMs' reputation for better generalization with sequential data, the GRU's simpler architecture led to fewer parameters, enabling more effective learning from our dataset. Additionally, the GRU demonstrated faster convergence during training, resulting in reduced training time and improved

TABLE 7. Detection result of the proposed approach on DCASE 2022 dataset.

Model	Feature	Training Evaluation					Testing Evaluation				
		Acc	Pre	Rec	F1	Training Time (s)	Acc	Pre	Rec	F1	Testing Time (s)
SVM	MS	0.4912	0.4271	0.4779	0.4181	2385.0636	0.4763	0.4228	0.4763	0.4102	704.0324
	MFCC	0.9215	0.9188	0.9177	0.9179	44.2827	0.8757	0.8755	0.87567	0.8733	19.127
	Both	0.9215	0.9188	0.9177	0.9179	248.7301	0.8756	0.8755	0.8757	0.8733	129.4131
DT	MS	0.7203	0.7152	0.7167	0.7158	465.8845	0.6597	0.6613	0.6597	0.6599	0.1193
	MFCC	0.8617	0.8603	0.8592	0.8595	91.5046	0.79	0.7897	0.79	0.7870	0.0135
	Both	0.8728	0.8704	0.8706	0.8705	581.6504	0.888	0.7934	0.7940	0.7919	0.136
XGBoost	MS	0.9624	0.9621	0.9627	0.9624	1659.9322	0.888	0.892	0.888	0.8875	0.376
	MFCC	0.9714	0.9707	0.9708	0.9707	201.691	0.904	0.8967	0.894	0.8920	0.057
	Both	0.9747	0.9742	0.9746	0.9744	1673.2859	0.9044	0.9029	0.9003	0.8988	0.386
RNN	MS	0.9139	0.9264	0.9176	0.9182	15945.6529	0.8514	0.872	0.8514	0.8492	33.7071
	MFCC	0.9762	0.9775	0.9773	0.9774	19949.5026	0.9326	0.9344	0.9326	0.932	32.8985
	Both	0.9725	0.9739	0.9736	0.9736	13058.4297	0.8945	0.9021	0.8945	0.8944	40.1997
LSTM	MS	0.9534	0.956	0.9559	0.9559	542.8892	0.9052	0.9089	0.9052	0.9049	4.8671
	MFCC	0.9588	0.9611	0.9611	0.9611	301.7337	0.9152	0.9167	0.9152	0.9152	2.4593
	Both	0.9642	0.9667	0.9664	0.9663	759.1009	0.9305	0.9325	0.9304	0.93	6.5355
GRU	MS	0.946	0.9505	0.949	0.9491	507.75	0.8983	0.9043	0.8983	0.8979	4.1606
	MFCC	0.9708	0.9722	0.9724	0.9723	222.8156	0.94	0.9394	0.94	0.9395	1.571
	Both	0.9659	0.9679	0.9678	0.9678	527.3554	0.931	0.9331	0.9309	0.9314	4.6605

performance. Given the specific requirements of our anomaly detection task, the GRU's ability to capture relevant temporal features with a shorter memory span was more effective than the LSTM's long-term memory capabilities.

- **Model Time Performance Evaluation:** For MIMII DG 2022 dataset, GRU and LSTM demonstrate moderate efficiency with MFCC features, with GRU achieving the fastest training time (160.09s) among neural networks. In contrast, RNN shows the poorest training performance, particularly with combined features (12,073.83s). While SVM exhibits fast training with MFCC (204.41s), it struggles with combined features (5,629.14s) and has notably long testing times (127-575s). Decision Trees emerge as the fastest in testing (0.0195-0.1966s), while XGBoost maintains consistent moderate performance in both phases. Notably, MFCC features consistently yield faster processing times compared to MS or combined features across all models, suggesting it's the most computationally efficient feature choice for this dataset.

For DCASE 2022 dataset, RNN demonstrates the longest training times across all feature types (13,058-19,949s), while SVM with MFCC features shows remarkably fast training (44.28s). Neural network models (LSTM, GRU) maintain moderate training times (222-759s), with MFCC features consistently

providing faster processing. XGBoost shows balanced performance but longer training times with MS and Both features (1,659-1,673s). For testing times, DT proves most efficient (0.0135-0.136s), while RNN has the longest testing times (32-40s). SVM shows notably high testing times with MS features (704s) but improves significantly with MFCC (19.12s). Overall, MFCC features consistently yield better computational efficiency across all models, with the combination of DT and MFCC features achieving the fastest testing time (0.0135s), while GRU with MFCC offers a good balance of both training (222.81s) and testing (1.571s) efficiency.

For the DCASE 2024 dataset, we employ the best-performing models from both ML and DL models based on the results of Tables 6 and 7. In ML, we utilize XGBoost with both feature sets, while in DL, we apply GRU with MFCC features. The evaluation results of our best models on this dataset are presented in Table 8.

About model time performance evaluation, GRU with MFCC demonstrates significantly faster training time (93.46s) compared to XGBoost with Both features (662.99s) - about 7 times faster. However, in testing phase, XGBoost shows better efficiency (0.297s) compared to GRU (0.7248s). Both models achieve exceptional accuracy metrics, with GRU reaching perfect scores in training and XGBoost showing slightly better testing performance, making the computational trade-offs particularly relevant for model selection in practical applications.

TABLE 8. Detection result of the proposed approach on DCASE 2024 dataset.

Model	Feature	Training Evaluation					Testing Evaluation				
		Acc	Pre	Rec	F1	Training Time (s)	Acc	Pre	Rec	F1	Testing Time (s)
XGBoost	Both	0.9989	0.9989	0.9989	0.9989	662.9903	0.9994	0.9994	0.9994	0.9994	0.297
GRU	MFCC	1	1	1	1	93.4617	0.9955	0.9956	0.9955	0.9955	0.7248

In summary, our approach highlights the effectiveness of employing a DL technique with the GRU model and carefully selected feature sets for highly accurate anomalous sound detection in large-scale industrial device sound datasets. These results emphasize the potential of our method to significantly enhance operational safety and efficiency in industrial environments.

B. MULTI-CLASS CLASSIFICATION RESULTS

We visualize the performance of the best models through confusion matrices, comparing XGBoost’s results with both feature combinations from ML approaches and GRU’s results with MFCC features from DL approaches, as illustrated in Figures 2 and 3. For other models performance result, we provide a more detailed analysis of our model’s performance using confusion matrices in Supplementary Figures 1 to 14.

Figure 2 illustrates the performance of XGBoost using combined features across three datasets: MIMII DG 2022, DCASE 2022, and DCASE 2024. For the MIMII DG 2022 dataset, the model demonstrates high accuracy with nearly perfect classification of “gearbox” and “fan,” but exhibits some confusion between “bearing” and “valve.” In contrast, the DCASE 2022 evaluation reveals generally strong performance, albeit with some cross-classification issues, particularly noticeable between “ToyCar” and “ToyTrain.” Notably, the DCASE 2024 dataset showcases exemplary results with perfect classification rates for all categories, reflecting exceptionally distinctive feature sets. Overall, XGBoost’s combined feature approach achieves robust performance, with minor challenges observed predominantly in the differentiation of mechanically similar components within the MIMII DG 2022 dataset.

Figure 3 presents confusion matrices evaluating the performance of the GRU model utilizing MFCC features across three datasets—MIMII DG 2022, DCASE 2022, and DCASE 2024. On the MIMII DG 2022 dataset, the model performs exceptionally well, achieving perfect classification for “bearing” and “valve,” and near-perfect for “fan” and “gearbox,” although there is some misclassification between “bearing” and “valve.” In contrast, the DCASE 2022 dataset shows varied success; while several components achieve high accuracy, there are noticeable cross-classifications, particularly involving “ToyCar” and “ToyTrain.” Finally, the DCASE 2024 dataset demonstrates outstanding performance with nearly flawless classification across all categories. This

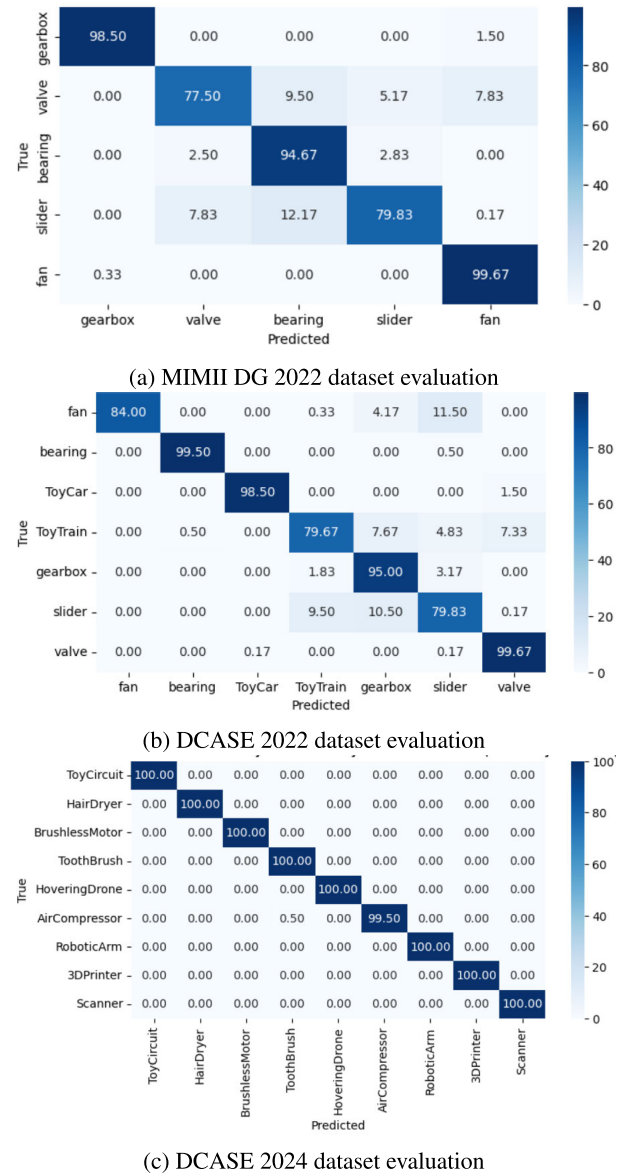


FIGURE 2. Confusion matrices showing XGBoost’s performance using combined features on test sets across three datasets (MIMII DG 2022, DCASE 2022, and DCASE 2024).

suggests that the GRU model, when paired with MFCC features, excels particularly in datasets where machine types exhibit clearer distinctions, while encountering some challenges in distinguishing between mechanically similar components.

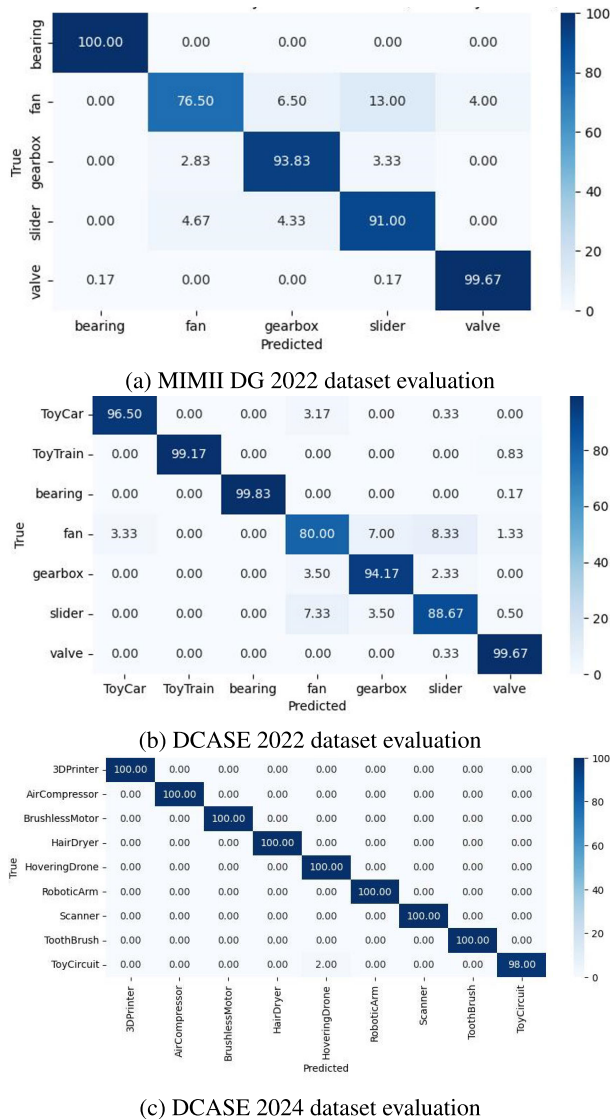


FIGURE 3. Confusion matrices showing GRU's performance using MFCC feature on test sets across three datasets (MIMII DG 2022, DCASE 2022, and DCASE 2024).

C. COMPARISON OF OUR PROPOSED METHOD WITH PREVIOUS APPROACHES

Table 9 compares our model with various prior models across different datasets (MIMII DG 2022 and DCASE 2022, 2024) based on accuracy, precision, recall, F1 score, training time, and testing time.

- For the MIMII DG 2022 dataset: The Transformer Decoder achieved an accuracy of 79.6%. MobileNetV2 had slightly lower accuracy at 75.83%, with a testing time of just 1 second. FFT and Sincet reached 84.94% accuracy but lacked other metrics like precision or recall.
- For the DCASE 2022 dataset: GRU+MFCC (ours) performed best with an accuracy of 94.2%, precision/recall/F1 scores all above 92, though it required significantly more training time compared to others. EfficientNetV2 was next with high accuracy (90.1%) but

no additional metrics provided. Tgram showed moderate performance with an accuracy of 68.29%.

- For the DCASE 2024 dataset: The GRU+MFCC model again led with an impressive 99.56% accuracy, maintaining high precision, recall, and F1 scores around 99.5%, along with notably low training and testing times. Autoencoder-based methods exhibited lower accuracies, ranging from 62.65% to 66%, without additional metrics reported. during training.

Overall, the GRU+MFCC model demonstrated superior performance consistently across both datasets, highlighting its effectiveness despite higher computational requirements.

D. IDENTIFICATION OF STRENGTHS AND LIMITATIONS OF THE PROPOSED APPROACH

This section analyzes the strengths and limitations of the proposed approach to anomalous sound detection in industrial environments. By examining the methodology's performance metrics and inherent characteristics, we aim to elucidate its efficacy and potential constraints in practical deployment scenarios.

Strengths:

- **High Accuracy:** The proposed approach consistently demonstrates high accuracy in ML and DL models across various datasets and feature sets. This indicates the robustness and effectiveness of the method in detecting anomalous sound events in industrial environments.
- **Feature Diversity:** By leveraging Mel-Spectrogram and MFCCs features, the approach captures a wide range of spectral characteristics inherent in industrial sound datasets. This feature diversity enhances the model's ability to discriminate between normal and abnormal sound instances.
- **Hybrid Approach:** Combining supervised ML and DL techniques offers a comprehensive solution for anomaly detection. This hybrid approach leverages the strengths of both paradigms, resulting in improved performance and generalization capabilities.
- **Model Flexibility:** The proposed approach allows for flexibility in model selection, enabling researchers to choose from various ML and DL algorithms based on their specific requirements and preferences.
- **Generalization:** While the proposed approach demonstrates strong performance across the evaluated datasets, its generalizability to unseen or highly diverse industrial environments remains to be explored.

Limitations:

- **Computational Complexity:** DL models, particularly those with recurrent architectures like LSTM and GRU, often require substantial computational resources for training and inference. This could pose challenges in real-time deployment or resource-constrained environments.
- **Interpretability:** While powerful, DL models often lack interpretability compared to traditional ML algorithms like decision trees. Understanding the inner workings of

TABLE 9. Comparison performance detection accuracy of our method with prior methods.

Dataset	Model	Accuracy	Precision	Recall	F1	Training time (s)	Testing time (s)
MIMII DG 2022	Transformer Decoder [72]	79.6	-	-	-	-	-
	MobileNetV2 [73]	75.83	-	-	-	-	1
	MobileNetV2, GMM [74]	56.22	-	-	-	-	-
	MPE, ResNet18 [75]	86.09	-	-	-	-	-
	FFT, Sincnet [76]	84.94	-	-	-	-	-
	GRU+MFCC (ours)	92.2	92.26	92.2	92.06	160.0866	1.0496
DCASE 2022	EfficientNetV2 [77]	90.1	-	-	-	-	-
	MobileNetV2 [78]	70.97	-	-	-	-	-
	GMM-based clustering [79]	85.54	-	-	-	-	-
	Tgram [80]	68.29	-	-	-	-	-
	Domain Generalization [81]	82.81	-	-	-	-	-
	DP-MAE [82]	74.10	-	-	-	-	-
	SSL [83]	86.0	-	-	-	-	-
	GRU+MFCC (ours)	94	93.94	94	93.95	222.8156	1.571
DCASE 2024	Autoencoder [65]	66	-	-	-	-	-
	AE [84]	62.65	-	-	-	-	-
	GRU+MFCC (Ours)	99.55	99.56	99.55	99.55	93.4617	0.7248

complex DL models may pose challenges in real-world applications where interpretability is crucial.

- **Model Tuning:** Achieving optimal performance with DL models typically requires extensive hyperparameter tuning and experimentation. This process can be time-consuming and resource-intensive, especially when dealing with large and complex datasets.

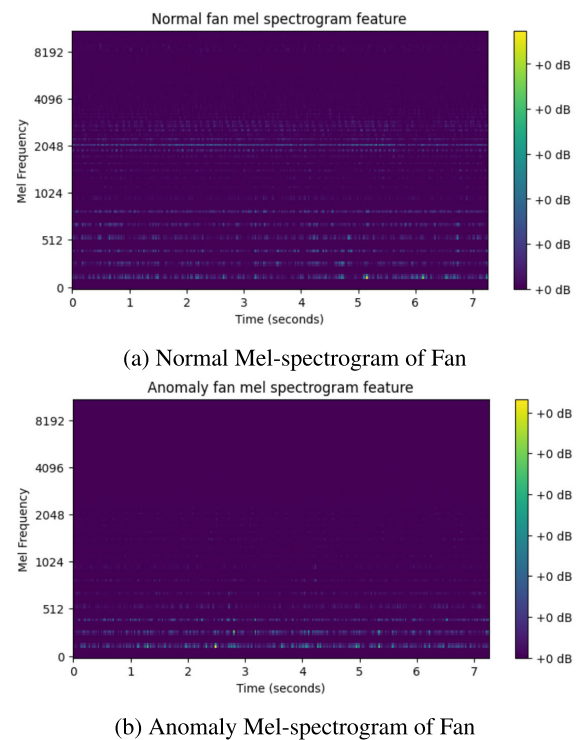
VI. ABLATION STUDY

This section presents ablation studies on: (1) the effect of high-frequency feature extraction and (2) the impact of our best model when utilizing different high-frequency features.

A. HIGH-FREQUENCY FEATURE EXTRACTION RESULTS

Our methodology extracts two primary high-frequency features from raw sound datasets: Mel-Spectrogram and MFCC. First, we present an example of the Mel-Spectrogram feature for a Fan device in Figure 4. Additionally, we illustrate the Mel-Spectrogram features extracted from all devices across the three datasets in Supplementary Figures 15-17. Several key observations emerge, as follows.

Figure 4 presents mel-spectrogram features for a fan device under normal and anomaly conditions from the MIMII DG 2022 dataset. The normal condition mel-spectrogram (a) shows consistent energy distribution across various frequencies over time, indicated by the presence of parallel horizontal lines throughout the spectrogram. In contrast, the anomaly condition mel-spectrogram (b) reveals disruptions in the regular pattern, characterized by sporadic bright spots and less structured frequency distributions. These visual cues highlight key differences between normal and

**FIGURE 4.** Mel-spectrogram feature of Fan device on MIMII DG 2022 dataset.

anomalous states, emphasizing the utility of mel-spectrograms in detecting deviations from typical operational behavior. Such discrepancies provide valuable information for developing anomaly detection systems capable of recognizing

early signs of malfunction or degradation in machinery health monitoring applications.

Second, we present an example of the Mel-Spectrogram feature for a Fan device in Figure 5. Additionally, we illustrate the MFCC features extracted from all devices across the three datasets in Supplementary Figures 18-20. Several key observations are highlighted below.

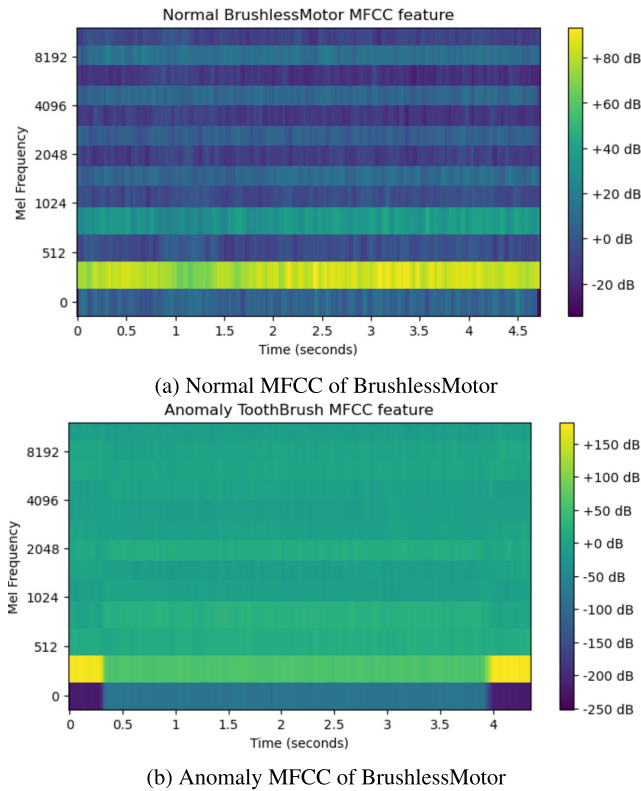


FIGURE 5. MFCC feature of BrushlessMotor device on DCASE 2024 dataset.

- **Discernible Differences:** MFCC values depict noticeable variations in MFCCs between normal and abnormal sound instances across various machines.
- **Machine-Specific Patterns:** MFCC values across the datasets reveal distinct patterns specific to each machine type, indicating unique spectral characteristics associated with individual devices.
- **Effective Discriminative Features:** The discernible variations in MFCC values between normal and abnormal sound instances underscore the efficacy of MFCCs as discriminative features for identifying anomalies within industrial sound datasets.
- **Potential for Anomaly Detection:** The observed differences in MFCC values highlight the potential utility of MFCCs in developing ML or DL-based anomaly detection systems for industrial equipment. This could facilitate the timely identification and mitigation of abnormal operating conditions.

Figure 5 presents MFCC features for BrushlessMotor devices under normal and anomaly conditions from the

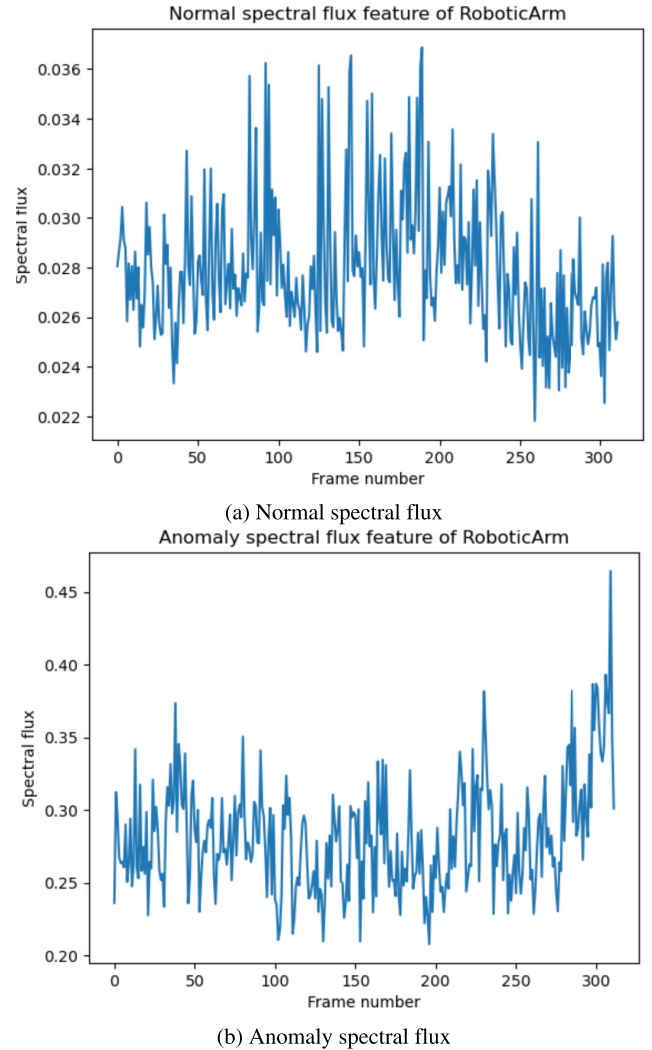


FIGURE 6. Spectral flux of RoboticArm device in normal and anomaly data of DCASE 2024 dataset.

DCASE 2024 dataset. The normal condition MFCC display (a) shows a stable pattern with alternating bands of blue and green, indicating consistent energy levels across various mel-frequency bins over time. In contrast, the anomaly condition MFCC (b) reveals irregularities with fewer defined bands and intermittent yellow regions, signaling deviations in the expected audio signature. These disparities suggest that abnormal operations lead to altered acoustic properties, crucial for building robust anomaly detection algorithms sensitive to changes in machine behavior. The color variations highlight the sensitivity of MFCCs in capturing subtle shifts in sound, thus offering critical insights into the health status of mechanical systems.

B. THE IMPACT OF DIFFERENT FEATURE EXTRACTION

To validate our high-frequency feature approach, we also extract another well-known high-frequency feature, spectral flux [84], [85]. First, we present the spectral flux data for the Robotic Arm device in Figure 6. Additionally, we visualize

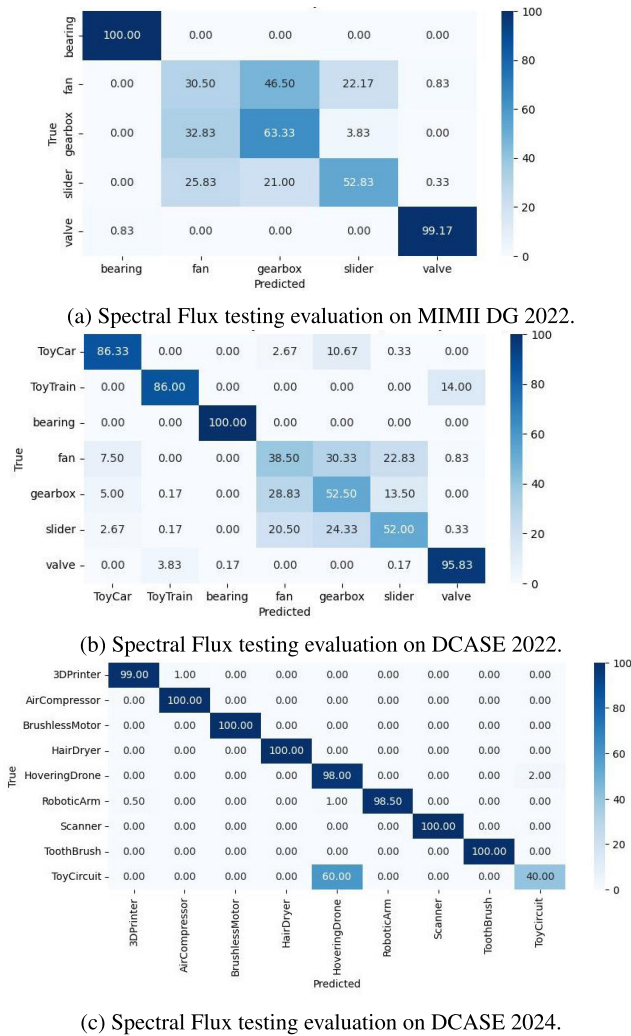


FIGURE 7. Confusion matrix results from the GRU model during the testing phase using the Spectral Flux feature.

the spectral flux results for all nine devices in Figure 21 in the Supplementary.

Figure 6 displays spectral flux features for the RoboticArm device under normal and anomaly conditions from the DCASE 2024 dataset. In the normal condition plot (a), the spectral flux fluctuates within a relatively narrow range, peaking occasionally but remaining consistently low overall. Conversely, the anomaly condition plot (b) reveals much larger fluctuations with frequent sharp peaks extending beyond 0.45, indicating irregularities in the signal. These differences highlight how anomalies can cause substantial deviations in spectral flux patterns, providing valuable insights for anomaly detection applications.

We evaluate our best model, GRU on the MIMII DG 2022 dataset and DCASE 2024 dataset, using the spectral flux feature. The results for GRU model with spectral flux are presented in Figure 7.

Figure 7 presents confusion matrix evaluations of the GRU model using the Spectral Flux feature across three

datasets: MIMII DG 2022, DCASE 2022, and DCASE 2024. The results showcase the model's ability to classify various machine components with high accuracy, particularly evident in the DCASE 2024 dataset, where it achieves near-perfect classification rates. However, within the MIMII DG 2022 and DCASE 2022 datasets, challenges persist, such as misclassifications between “fan” and “gearbox,” and difficulties in distinguishing “ToyCar” from “ToyTrain.” These limitations suggest that the Spectral Flux feature may struggle with components exhibiting similar spectral characteristics, highlighting areas for potential improvements in feature extraction techniques or model enhancements to enhance discrimination between closely related acoustic signals.

VII. CONCLUSION

In conclusion, our study has demonstrated the effectiveness of supervised learning approaches, particularly XGBoost in the ML realm, and GRU in the DL domain, for anomalous sound detection in industrial environments. Through rigorous experimentation on diverse datasets, including the MIMII DG, DCASE 2022, and DCASE 2024 datasets, our findings indicate clear trends in model performance.

In the ML approach, XGBoost consistently outperforms other ML models, showcasing its robustness and efficiency in capturing complex patterns within the data. Conversely, within the DL approach, GRU emerges as the top performer, demonstrating its ability to capture temporal dependencies in sound signals effectively. Moreover, when comparing the ML and DL approaches, our results highlight the superiority of GRU with Mel-Spectrogram features over XGBoost with both Mel-Spectrogram and MFCC features across three datasets. This underscores the importance of selecting appropriate features and model architectures tailored to the specific characteristics of the data.

Although our study has provided valuable information on the efficacy of supervised learning techniques for anomalous sound detection, certain limitations should be acknowledged. Future research efforts could explore ensemble learning strategies, incorporate domain-specific knowledge, and address issues related to model interpretability and false positives in real-world applications. Our study contributes to the growing literature on anomaly detection in industrial settings, offering practical implications for developing robust and accurate detection systems. By leveraging the strengths of both ML and DL approaches, we pave the way for more effective solutions in industrial sound monitoring and anomaly detection.

REFERENCES

- [1] P. Muchiri, L. Pintelon, L. Gelders, and H. Martin, “Development of maintenance function performance measurement framework and indicators,” *Int. J. Prod. Econ.*, vol. 131, no. 1, pp. 295–302, May 2011.
- [2] A. Diez-Oliván, J. Del Ser, D. Galar, and B. Sierra, “Data fusion and machine learning for industrial prognosis: Trends and perspectives towards industry 4.0,” *Inf. Fusion*, vol. 50, pp. 92–111, Oct. 2019.

- [3] A. Abbas, "AI for predictive maintenance in industrial systems," *Int. J. Adv. Eng. Technol. Innov.*, vol. 1, no. 1, pp. 31–51, 2024.
- [4] S. Langerica, C. Rüffelmacher, and F. Núñez, "An industrial Internet application for real-time fault diagnosis in industrial motors," *IEEE Trans. Autom. Sci. Eng.*, vol. 17, no. 1, pp. 284–295, Jan. 2020, doi: [10.1109/TASE.2019.2913628](https://doi.org/10.1109/TASE.2019.2913628).
- [5] S. Gawde, S. Patil, S. Kumar, P. Kamat, K. Kotecha, and A. Abraham, "Multi-fault diagnosis of industrial rotating machines using data-driven approach: A review of two decades of research," *Eng. Appl. Artif. Intell.*, vol. 123, Aug. 2023, Art. no. 106139.
- [6] D. Pachori, R. K. Tripathy, and T. K. Jain, "Detection of atrial fibrillation from PPG sensor data using variational mode decomposition," *IEEE Sensors Lett.*, vol. 8, no. 3, pp. 1–4, Mar. 2024.
- [7] D. Pachori and T. Kumar Gandhi, "FBSE-based approach for discriminating seizure and normal EEG signals," *IEEE Sensors Lett.*, vol. 8, no. 12, pp. 1–4, Dec. 2024.
- [8] D. Pachori and T. K. Gandhi, "Automated human emotion recognition system using TQWT-based EEG subbands," *IEEE Sensors Lett.*, vol. 8, no. 12, pp. 1–4, Dec. 2024.
- [9] D. Pachori, S. Dash, R. K. Tripathy, and T. K. Jain, "Introduction to cardiovascular signals and automated systems," *Signal Process. Driven Mach. Learn. Techn. Cardiovascular Data Process.*, vol. 2024, pp. 1–17, Jun. 2024.
- [10] S. Delvecchio, P. Bonfiglio, and F. Pompili, "Vibro-acoustic condition monitoring of internal combustion engines: A critical review of existing techniques," *Mech. Syst. Signal Process.*, vol. 99, pp. 661–683, Jan. 2018.
- [11] M. H. Mohd Ghazali and W. Rahiman, "Vibration analysis for machine monitoring and diagnosis: A systematic review," *Shock Vibrat.*, vol. 2021, no. 1, pp. 1–25, Jan. 2021.
- [12] N. Choudhry, J. Abawajy, S. Huda, and I. Rao, "A comprehensive survey of machine learning methods for surveillance videos anomaly detection," *IEEE Access*, vol. 11, pp. 114680–114713, 2023, doi: [10.1109/access.2023.3321800](https://doi.org/10.1109/access.2023.3321800).
- [13] P. Yan, A. Abdulkadir, P.-P. Luley, M. Rosenthal, G. A. Schatte, B. F. Grewe, and T. Stadelmann, "A comprehensive survey of deep transfer learning for anomaly detection in industrial time series: Methods, applications, and directions," *IEEE Access*, vol. 12, pp. 3768–3789, 2024, doi: [10.1109/ACCESS.2023.3349132](https://doi.org/10.1109/ACCESS.2023.3349132).
- [14] L. You, W. Fan, Z. Li, Y. Liang, M. Fang, and J. Wang, "A fault diagnosis model for rotating machinery using VWC and MSFLA-SVM based on vibration signal analysis," *Shock Vibrat.*, vol. 2019, no. 1, Jan. 2019, Art. no. 1908485.
- [15] J. Saari, D. Strömbergsson, J. Lundberg, and A. Thomson, "Detection and identification of windmill bearing faults using a one-class support vector machine (SVM)," *Measurement*, vol. 137, pp. 287–301, Apr. 2019.
- [16] A. Kafeel, S. Aziz, M. Awais, M. A. Khan, K. Afaq, S. A. Idris, H. Alshazly, and S. M. Mostafa, "An expert system for rotating machine fault detection using vibration signal analysis," *Sensors*, vol. 21, no. 22, p. 7587, Nov. 2021.
- [17] K. Vos, Z. Peng, C. Jenkins, M. R. Shahriar, P. Borghesani, and W. Wang, "Vibration-based anomaly detection using LSTM/SVM approaches," *Mech. Syst. Signal Process.*, vol. 169, Apr. 2022, Art. no. 108752.
- [18] P. Jirapipattanaporn and W. Lawanont, "Development of anomaly detection model for welding classification using arc sound," in *Proc. 14th Int. Conf. Knowl. Smart Technol. (KST)*, Chon buri, Thailand, Jan. 2022, pp. 57–62, doi: [10.1109/KST53302.2022.9729058](https://doi.org/10.1109/KST53302.2022.9729058).
- [19] D. L. Aguilar, M. A. Medina-Pérez, O. Loyola-González, K.-K. R. Choo, and E. Bucheli-Susarrey, "Towards an interpretable autoencoder: A decision-tree-based autoencoder and its application in anomaly detection," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 2, pp. 1048–1059, Apr. 2023, doi: [10.1109/TDSC.2022.3148331](https://doi.org/10.1109/TDSC.2022.3148331).
- [20] S. Naseer, Y. Saleem, S. Khalid, M. K. Bashir, J. Han, M. M. Iqbal, and K. Han, "Enhanced network anomaly detection based on deep neural networks," *IEEE Access*, vol. 6, pp. 48231–48246, 2018, doi: [10.1109/ACCESS.2018.2863036](https://doi.org/10.1109/ACCESS.2018.2863036).
- [21] H. Xu, P. Song, and B. Liu, "A vibration signal anomaly detection method based on frequency component clustering and isolated forest algorithm," in *Proc. IEEE 2nd Int. Conf. Autom., Electron. Electr. Eng. (AUTEEE)*, Shenyang, China, Nov. 2019, pp. 49–53, doi: [10.1109/AUTEEE48671.2019.9033363](https://doi.org/10.1109/AUTEEE48671.2019.9033363).
- [22] M. Baydoun, L. Safatly, H. Ghaziri, and A. El Hajj, "Analysis of heart sound anomalies using ensemble learning," *Biomed. Signal Process. Control*, vol. 62, Sep. 2020, Art. no. 102019.
- [23] P. D. Rosero-Montalvo, "Anomaly detection in industrial air conditioners in hangars with aircraft spare parts," *IEEE Sensors Lett.*, vol. 8, no. 4, Apr. 2024, Art. no. 5501404, doi: [10.1109/LESENS.2024.3374260](https://doi.org/10.1109/LESENS.2024.3374260).
- [24] P. Harar, Z. Galaz, J. B. Alonso-Hernandez, J. Mekyska, R. Burget, and Z. Smekal, "Towards robust voice pathology detection: Investigation of supervised deep learning, gradient boosting, and anomaly detection approaches across four databases," *Neural Comput. Appl.*, vol. 32, no. 20, pp. 15747–15757, Oct. 2020.
- [25] E. Akca, T. Özçay, Y. Dinç, N. Yalçı, S. E. Taşabat, M. A. Varol, B. Kayı, M. Y. Ögtcen, and B. Öztürk, "CNN based ensemble approach for malfunction detection from machine sounds," *Eur. J. Res. Develop.*, vol. 2, no. 2, pp. 411–420, Jun. 2022.
- [26] A. Rath, D. Mishra, G. Panda, and M. Pal, "Development and assessment of machine learning based heart disease detection using imbalanced heart sound signal," *Biomed. Signal Process. Control*, vol. 76, Jul. 2022, Art. no. 103730.
- [27] R. K. Tripathy, S. Dash, A. Rath, G. Panda, and R. B. Pachori, "Automated detection of pulmonary diseases from lung sound signals using fixed-boundary-based empirical wavelet transform," *IEEE Sensors Lett.*, vol. 6, no. 5, pp. 1–4, May 2022, doi: [10.1109/LESENS.2022.3167121](https://doi.org/10.1109/LESENS.2022.3167121).
- [28] M. K. Hooshmand, M. D. Huchaiha, A. R. Alzighaibi, H. Hashim, E.-S. Atlam, and I. Gad, "Robust network anomaly detection using ensemble learning approach and explainable artificial intelligence (XAI)," *Alexandria Eng. J.*, vol. 94, pp. 120–130, May 2024.
- [29] V. G. Sujadevi, K. P. Soman, R. Vinayakumar, and A. U. P. Sankar, "Anomaly detection in phonocardiogram employing DL," in *Proc. Int. Conf. Comput. Intell. Data Mining CIDM*. Singapore: Springer, 2019, pp. 525–534.
- [30] S. Latif, M. Usman, R. Rana, and J. Qadir, "Phonocardiographic sensing using deep learning for abnormal heartbeat detection," *IEEE Sensors J.*, vol. 18, no. 22, pp. 9393–9400, Nov. 2018, doi: [10.1109/jsen.2018.2870759](https://doi.org/10.1109/jsen.2018.2870759).
- [31] Y. Park and I. D. Yun, "Fast adaptive RNN encoder-decoder for anomaly detection in SMD assembly machine," *Sensors*, vol. 18, no. 10, p. 3573, Oct. 2018.
- [32] M. Canizo, I. Triguero, A. Conde, and E. Onieva, "Multi-head CNN-RNN for multi-time series anomaly detection: An industrial case study," *Neurocomputing*, vol. 363, pp. 246–260, Oct. 2019.
- [33] A. Manzoor, Q. Pan, H. J. Khan, S. Siddeeq, H. M. A. Bhatti, and M. A. Wedagu, "Analysis and detection of lung sounds anomalies based on NMA-RNN," in *Proc. IEEE Int. Conf. Bioinf. Biomed. (BIBM)*, Dec. 2020, pp. 2498–2504, doi: [10.1109/BIBM49941.2020.9313197](https://doi.org/10.1109/BIBM49941.2020.9313197).
- [34] Z. Li, J. Li, Y. Wang, and K. Wang, "A deep learning approach for anomaly detection based on SAE and LSTM in mechanical equipment," *Int. J. Adv. Manuf. Technol.*, vol. 103, nos. 1–4, pp. 499–510, Jul. 2019.
- [35] O. I. Provotar, Y. M. Linder, and M. M. Veres, "Unsupervised anomaly detection in time series using LSTM-based autoencoders," in *Proc. IEEE Int. Conf. Adv. Trends Inf. Theory (ATIT)*, Kyiv, Ukraine, Dec. 2019, pp. 513–517, doi: [10.1109/ATIT49449.2019.9030505](https://doi.org/10.1109/ATIT49449.2019.9030505).
- [36] P. Becker, C. Roth, A. Roennau, and R. Dillmann, "Acoustic anomaly detection in additive manufacturing with long short-term memory neural networks," in *Proc. IEEE 7th Int. Conf. Ind. Eng. Appl. (ICIEA)*, Bangkok, Thailand, Apr. 2020, pp. 921–926, doi: [10.1109/ICIEA49774.2020.9102002](https://doi.org/10.1109/ICIEA49774.2020.9102002).
- [37] T. Tran, N. Truong Pham, and J. Lundgren, "Detecting drill failure in the small short-sound drill dataset," 2021, *arXiv:2108.11089*.
- [38] C. Wall, L. Zhang, Y. Yu, and K. Mistry, "Deep recurrent neural networks with attention mechanisms for respiratory anomaly classification," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Shenzhen, China, Jul. 2021, pp. 1–8, doi: [10.1109/IJCNN52387.2021.9533966](https://doi.org/10.1109/IJCNN52387.2021.9533966).
- [39] H. Ahn and I. Yeo, "Deep-Learning-Based approach to anomaly detection techniques for large acoustic data in machine operation," *Sensors*, vol. 21, no. 16, p. 5446, Aug. 2021.
- [40] L. Z. Yong and H. Nugroho, "Acoustic anomaly detection of mechanical failure: Time-distributed CNN-RNN DL models," in *Control, Instrumentation and Mechatronics: Theory and Practice*. Singapore: Springer, 2022, pp. 662–672.
- [41] P. Mobtahej, X. Zhang, M. Hamidi, and J. Zhang, "An LSTM-autoencoder architecture for anomaly detection applied on compressors audio data," *Comput. Math. Methods*, vol. 2022, pp. 1–22, Sep. 2022.
- [42] A. Qurhobi and R. Maskeliūnas, "Deep learning and acoustic approach for mechanical failure detection in industrial machinery," *J. Phys., Conf. Ser.*, vol. 2673, no. 1, Dec. 2023, Art. no. 012032.

- [43] F. Qiu, S. Chen, D. Jin, and A. Lu, "Fan sound fault detection algorithm based on the fusion of convolutional neural network and gated recurrent unit network," in *Proc. 6th Int. Conf. Adv. Electron. Mater., Comput., Softw. Eng. (AEMCSE)*, Aug. 2023, p. 13.
- [44] H. Yadav, P. Shah, N. Gandhi, T. Vyas, A. Nair, S. Desai, and L. Gohil, "Cnn and bidirectional gru-based heartbeat sound classification architecture for elderly people," *Mathematics*, vol. 11, no. 6, p. 1365, 2023.
- [45] R. Müller, F. Ritz, S. Illium, and C. Linnhoff-Popien, "Acoustic anomaly detection for machine sounds based on image transfer learning," 2020, *arXiv:2006.03429*.
- [46] R. Müller, S. Illium, and C. Linnhoff-Popien, "Deep recurrent interpolation networks for anomalous sound detection," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2021, pp. 1–7.
- [47] T. Tran and J. Lundgren, "Drill fault diagnosis based on the scalogram and mel spectrogram of sound signals using artificial intelligence," *IEEE Access*, vol. 8, pp. 203655–203666, 2020, doi: [10.1109/ACCESS.2020.3036769](https://doi.org/10.1109/ACCESS.2020.3036769).
- [48] Y. Deng, A. Jiang, Y. Duan, J. Ma, X. Chen, J. Liu, P. Fan, C. Lu, and W.-Q. Zhang, "Ensemble of multiple anomalous sound detectors," in *Proc. Detection and Classification of Acoustic Scenes and Events*, Nancy, France, Nov. 2022. [Online]. Available: https://dcase.community/documents/workshop2022/proceedings/DCASE2022Workshop_Deng_4.pdf
- [49] L. Melgar-García, M. Hosseini, AND A. Troncoso, "Identification of anomalies in urban sound data with autoencoders," in *Proc. Int. Conf. Hybrid Artif. Intell. Syst.* Cham, Switzerland: Springer, 2023, pp. 27–38.
- [50] A. Ferraro, A. Galli, V. L. Gatta, V. Moscato, M. Postiglione, G. Sperl, and F. Moscato, "Unsupervised anomaly detection in predictive maintenance using sound data," in *Proc. SEBD*, Jul. 2023, pp. 449–458.
- [51] Y. U. N. Eunsun and M. Jeong, "Enhancing automotive motor anomaly detection with mel spectrogram image transformation," in *Proc. ICCV*, 2023, pp. 279–280.
- [52] F. Zou, X. Li, Y. Li, S. Sang, M. Jiang, and H. Zhang, "GOL-SFSTS based few-shot learning mechanical anomaly detection using multi-channel audio signal," *Knowl.-Based Syst.*, vol. 284, Jan. 2024, Art. no. 111204.
- [53] J. Liu, Y. Yin, H. Jiang, H. Kan, Z. Zhang, P. Chen, B. Zhu, and Z. Wang, "Bowel sound detection based on MFCC feature and LSTM neural network," in *Proc. IEEE Biomed. Circuits Syst. Conf. (BioCAS)*, Cleveland, OH, USA, Oct. 2018, pp. 1–4, doi: [10.1109/BIOCAS.2018.8584723](https://doi.org/10.1109/BIOCAS.2018.8584723).
- [54] C. Lubis and F. Gondawijaya, "Heart sound diagnose system with BFCC, MFCC, and backpropagation neural network," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 508, May 2019, Art. no. 012119.
- [55] M. Deng, T. Meng, J. Cao, S. Wang, J. Zhang, and H. Fan, "Heart sound classification based on improved MFCC features and convolutional recurrent neural networks," *Neural Netw.*, vol. 130, pp. 22–32, Oct. 2020.
- [56] F. Lyu, Z. Yang, L. Wang, Q. Zhou, S. Hou, and X. Li, "Behavior anomaly detection fused with features of mel frequency cepstrum coefficients," in *Proc. Int. Conf. Sens., Meas. Data Analytics era Artif. Intell. (ICSMD)*, Xi'an, China, Oct. 2020, pp. 89–93, doi: [10.1109/ICSMD50554.2020.9261684](https://doi.org/10.1109/ICSMD50554.2020.9261684).
- [57] A.-U. Rehman, H. S. Ullah, H. Farooq, M. S. Khan, T. Mahmood, and H. O. A. Khan, "Multi-modal anomaly detection by using audio and visual cues," *IEEE Access*, vol. 9, pp. 30587–30603, 2021, doi: [10.1109/ACCESS.2021.3059519](https://doi.org/10.1109/ACCESS.2021.3059519).
- [58] P. Mobtahej, X. Zhang, M. Hamidi, and J. Zhang, "Deep learning-based anomaly detection for compressors using audio data," in *Proc. Annu. Rel. Maintainability Symp. (RAMS)*, Orlando, FL, USA, May 2021, pp. 1–7, doi: [10.1109/rams48097.2021.9605720](https://doi.org/10.1109/rams48097.2021.9605720).
- [59] F. Hua and L. Li, "Sound anomaly detection of industrial products based on MFCC fusion short-time energy feature extraction," in *Proc. IEEE Conf. Telecommun., Opt. Comput. Sci. (TOCS)*, Dalian, China, Dec. 2022, pp. 861–864, doi: [10.1109/TOCS56154.2022.10016076](https://doi.org/10.1109/TOCS56154.2022.10016076).
- [60] A. Abbasi, A. R. R. Javed, A. Yasin, Z. Jalil, N. Kryvinska, and U. Tariq, "A large-scale benchmark dataset for anomaly detection and rare event classification for audio forensics," *IEEE Access*, vol. 10, pp. 38885–38894, 2022, doi: [10.1109/ACCESS.2022.3166602](https://doi.org/10.1109/ACCESS.2022.3166602).
- [61] M. Chuphal, K. Singh, A. Bisht, V. Sharma, S. Awasthi, and S. Vats, "Anomaly detection in electro-mechanical devices using MFCC," in *Proc. 2nd Int. Conf. Disruptive Technol. (ICDT)*, Mar. 2024, pp. 986–991, doi: [10.1109/ICDT61202.2024.10489646](https://doi.org/10.1109/ICDT61202.2024.10489646).
- [62] K. Dohi, T. Nishida, H. Purohit, R. Tanabe, T. Endo, M. Yamamoto, Y. Nikaido, and Y. Kawaguchi, "MIMI DG: Sound dataset for malfunctioning industrial machine investigation and inspection for domain generalization task," 2022, *arXiv:2205.13879*.
- [63] K. Dohi, K. Imoto, N. Harada, D. Niizumi, Y. Koizumi, T. Nishida, H. Purohit, T. Endo, M. Yamamoto, and Y. Kawaguchi, "Description and discussion on DCASE 2022 challenge task 2: Unsupervised anomalous sound detection for machine condition monitoring applying domain generalization techniques," 2022, *arXiv:2206.05876*.
- [64] T. Nishida, N. Harada, D. Niizumi, D. Albertini, R. Sannino, S. Pradolini, F. Augusti, K. Imoto, K. Dohi, H. Purohit, T. Endo, and Y. Kawaguchi, "Description and discussion on DCASE 2024 challenge task 2: First-shot unsupervised anomalous sound detection for machine condition monitoring," 2024, *arXiv:2406.07250*.
- [65] P. Wang, J. Wang, and L. F. Li, "Machine learning model based on non-convex penalized huberized-SVM," *J. Electron. Sci. Technol.*, vol. 22, no. 1, 2024, Art. no. 100246.
- [66] M. Zenga and C. Colapinto, "Decision tree for classification and forecasting," in *Engineering Mathematics and Artificial Intelligence*. Boca Raton, FL, USA: CRC Press, 2023, pp. 53–82.
- [67] E. K. Sahin, "Assessing the predictive capability of ensemble tree methods for landslide susceptibility mapping using XGBoost, gradient boosting machine, and random forest," *Social Netw. Appl. Sci.*, vol. 2, no. 7, p. 1308, Jul. 2020.
- [68] A. Sherstinsky, "Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network," *Phys. D, Nonlinear Phenomena*, vol. 404, Mar. 2020, Art. no. 132306.
- [69] J. H. K. Larcher, S. F. Stefenon, L. D. S. Coelho, and V. C. Mariani, "Enhanced multi-step streamflow series forecasting using hybrid signal decomposition and optimized reservoir computing models," *Expert Syst. Appl.*, vol. 255, Dec. 2024, Art. no. 124856.
- [70] Y. Yu, X. Si, C. Hu, and J. Zhang, "A review of recurrent neural networks: LSTM cells and network architectures," *Neural Comput.*, vol. 31, no. 7, pp. 1235–1270, Jul. 2019.
- [71] S. T. Y. Kawaguchi, T. Nishida, K. Imoto, Y. Okamoto, K. Dohi, and T. Endo, "Audio-change captioning to explain machine-sound anomalies," in *Proc. DCASE Workshop*, Tampere, Finland, Sep. 2023. [Online]. Available: https://dcase.community/documents/workshop2023/proceedings/DCASE2023Workshop_Tsubaki_8.pdf
- [72] D. Abdelhady, Y. Abdelrahman, and H. Othman, "Outlier exposure-based anomalous sound detection using DL," in *Proc. 11th Int. Conf. Intell. Comput. Inf. Syst. (ICICIS)*, Apr. 2023, pp. 73–79.
- [73] N. Amer and H. Othman, "Time stretching and pitch shifting for anomalous sound detection in fan sound signal using MobileNetV2," in *Proc. 11th Int. Conf. Intell. Comput. Inf. Syst. (ICICIS)*, Nov. 2023, pp. 80–86, doi: [10.1109/ICICIS58388.2023.10391190](https://doi.org/10.1109/ICICIS58388.2023.10391190).
- [74] X.-M. Zeng, Y. Song, I. McLoughlin, L. Liu, and L.-R. Dai, "Robust prototype learning for anomalous sound detection," in *Proc. INTERSPEECH*, Aug. 2023, pp. 261–265.
- [75] J. Geng, Y. Liao, L. Guo, X. Ma, and G. Wang, "Motor equipment fault diagnostics via multi-feature acoustic signal fusion," in *Proc. CAA Symp. Fault Detection, Supervision Saf. Tech. Processes (SAFEPROCESS)*, Sep. 2023, pp. 1–6.
- [76] J.-Y. Kuo, C.-Y. Hsu, P.-F. Wang, H.-C. Lin, and Z.-G. Nie, "Constructing condition monitoring model of harmonic drive," *Appl. Sci.*, vol. 12, no. 19, p. 9415, Sep. 2022, doi: [10.3390/app12199415](https://doi.org/10.3390/app12199415).
- [77] S. Venkatesh, G. Wichern, A. Subramanian, and J. L. Roux, "Disentangled surrogate task learning for improved domain generalization in unsupervised anomalous sound detection," in *Proc. DCASE*, 2022. [Online]. Available: https://dcase.community/documents/challenge2022/technical_reports/DCASE2022_Venkatesh_99_t2.pdf
- [78] F. Xiao, Y. Liu, Y. Wei, J. Guan, Q. Zhu, T. Zheng, and J. Han, "The DCASE2022 challenge task 2 system: Anomalous sound detection with self-supervised attribute classification and GMM-based clustering," Nancy, France, Tech. Rep., Nov. 2022.
- [79] S. Wang, J. Du, and Y. Wang, "Machine anomalous sound detection based on self-supervised classification," in *Proc. Asia Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, Nov. 2022, pp. 449–454.
- [80] I. Kuroyanagi, T. Hayashi, K. Takeda, and T. Toda, "Two-stage anomalous sound detection systems using domain generalization and specialization techniques," in *Proc. DCASE*, 2022. [Online]. Available: https://dcase.community/documents/challenge2022/technical_reports/DCASE2022_Kuroyanagi_11_t2.pdf
- [81] Z.-L. Liu, Y. Song, X.-M. Zeng, L.-R. Dai, and I. McLoughlin, "DP-MAE: A dual-path masked autoencoder based self-supervised learning method for anomalous sound detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 1481–1485.

- [82] K. Wilkinghoff, "Self-supervised learning for anomalous sound detection," in *Proc. ICASSP - IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 276–280.
- [83] S. Wang, G. Zhong, Q. Wang, and J. Du, "Hybrid anomaly detection approach for DCASE 2024 task 2," Nancy, France, Tech. Rep., Nov. 2022.
- [84] S. Lee, J. Kim, and I. Lee, "Speech/Audio signal classification using spectral flux pattern recognition," in *Proc. IEEE Workshop Signal Process. Syst.*, Oct. 2012, pp. 232–236.
- [85] B. Boashash, N. A. Khan, and T. Ben-Jabeur, "Time–frequency features for pattern recognition using high-resolution TFDs: A tutorial review," *Digit. Signal Process.*, vol. 40, pp. 1–30, May 2015.



JIWON YUN received the bachelor's degree in computer engineering from Pusan National University, South Korea, in 2024, where she is currently pursuing the master's degree in computer science. Her current research interests include AI, NLP, and generative AI.



THI-THU-HUONG LE received the bachelor's degree from the Hung Yen University of Technology and Education (HYUTE), Vietnam, in 2007, the master's degree in technology with HUST, in 2013, and the Ph.D. degree from Pusan National University (PNU), South Korea, in 2020. She is a Research Professor with the Blockchain Platform Research Center, PNU. With a robust academic background, she has accumulated a wealth of experience. In 2020, she was a Postdoctoral Researcher with PNU, for three years. Before this, she was a Lecturer with HYUTE, for seven years. Her professional journey has been enriched by active participation in ML projects, covering areas such as NILM, IDS, AI Industry 4.0, AI security, and DL-based CFD. Her research interests include ML, DL, data analysis, explainable AI, AI security, and signal processing.



ANDRO APRILA ADIPUTRA received the bachelor's degree in electronic engineering from the Electronic Engineering Polytechnics Institute of Surabaya, Indonesia, in 2022. He is a dedicated Researcher and a Graduate Student, specializing in computer engineering with Pusan National University, South Korea. His research interests span a broad array of topics within the field of artificial intelligence, including ML, computer vision, and large language models (LLMs).



HOWON KIM (Member, IEEE) received the bachelor's degree from Kyungpook National University (KNU), and the Ph.D. degree from Pohang University of Science and Technology (POSTECH). He is currently a Professor with the Department of Electrical and Computer Engineering, PNU. His role extends beyond academia, encompassing leadership positions as the Chief of the Energy Internet of Things (IoT) with the IT Research Center (ITRC) and the Chief of the Information Security Education Center (ISEC), PNU. Before his tenure with PNU, his career journey included a significant stint with the Electronics and Telecommunications Research Institute (ETRI), where he was a Team Leader for a decade, since in December 1998. His pursuit of knowledge led him to Ruhr-University Bochum, Germany, where he undertook the role of a Postdoctoral Researcher with the Chair for Communication Security Group (COSY), from July 2003 to June 2004.

...